AID Price Paid

GROUP 39:
TIAGO DEANE (103811)
AFONSO MATOS (103479)

TASK 2 - Create and load initial database

```sql
SET NAMES utf8;

DROP DATABASE IF EXISTS pp_database;

CREATE DATABASE IF NOT EXISTS pp_database DEFAULT CHARACTER SET utf8;

USE pp_database;

CREATE TABLE IF NOT EXISTS pp_table (
    transactionid CHAR(38)       ,
    price         INT            ,
    transferdate  DATETIME       ,
    postcode      VARCHAR(10)    ,
    propertytype  CHAR(1)        ,
    newbuild      CHAR(1)        ,
    tenure        CHAR(1)        ,
    paon          VARCHAR(100)   ,
    saon          VARCHAR(50)    ,
    streetname    VARCHAR(50)    ,
    locality      VARCHAR(50)    ,
    city          VARCHAR(50)    ,
    district      VARCHAR(50)    ,
    county        VARCHAR(50)    ,
    recordstatus  CHAR(1)        ,
    category      CHAR(1)        ,
    PRIMARY KEY (transactionid)
);

LOAD DATA LOCAL INFILE '/home/aid/Downloads/pp-2025.csv'
INTO TABLE pp_table
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
(transactionid, price, transferdate, postcode, propertytype, newbuild, tenure, paon, saon, streetname, locality, city, district, county, recordstatus, category);
```

TASK 3

```sql
SELECT
    city,
    DATE(transferdate) AS sale_day,
    SUM(price) AS total_sales
FROM pp_table
GROUP BY city, sale_day
ORDER BY total_sales DESC
LIMIT 20;
```

```
mysql> SELECT
    ->     city,
    ->     DATE(transferdate) AS sale_day,
    ->     SUM(price) AS total_sales
    -> FROM pp_table
    -> GROUP BY city, sale_day
    -> ORDER BY total_sales DESC
    -> LIMIT 20;
+--------+------------+-------------+
| city   | sale_day   | total_sales |
+--------+------------+-------------+
| LONDON | 2025-03-28 |   990087053 |
| LONDON | 2025-03-31 |   939942604 |
| LONDON | 2025-02-28 |   531966671 |
| LONDON | 2025-01-31 |   505244016 |
| LONDON | 2025-03-21 |   461877869 |
| LONDON | 2025-02-21 |   429393607 |
| LONDON | 2025-03-27 |   385810744 |
| LONDON | 2025-02-14 |   366606613 |
| LONDON | 2025-03-14 |   333689232 |
| LONDON | 2025-02-07 |   316537774 |
| LONDON | 2025-03-07 |   282818983 |
| LONDON | 2025-03-26 |   282114758 |
| LONDON | 2025-06-27 |   273537186 |
| LONDON | 2025-01-10 |   267351750 |
| LONDON | 2025-01-17 |   264291641 |
| LONDON | 2025-01-24 |   262489417 |
| LONDON | 2025-05-30 |   252783256 |
| LONDON | 2025-06-13 |   228973893 |
| LONDON | 2025-01-21 |   213773035 |
| LONDON | 2025-01-15 |   207573086 |
+--------+------------+-------------+
20 rows in set (0.73 sec)
```

TASK 4

Create the data warehouse

```sql
SET NAMES utf8;

DROP DATABASE IF EXISTS pp_star_database;

CREATE DATABASE IF NOT EXISTS pp_star_database DEFAULT CHARACTER SET utf8;

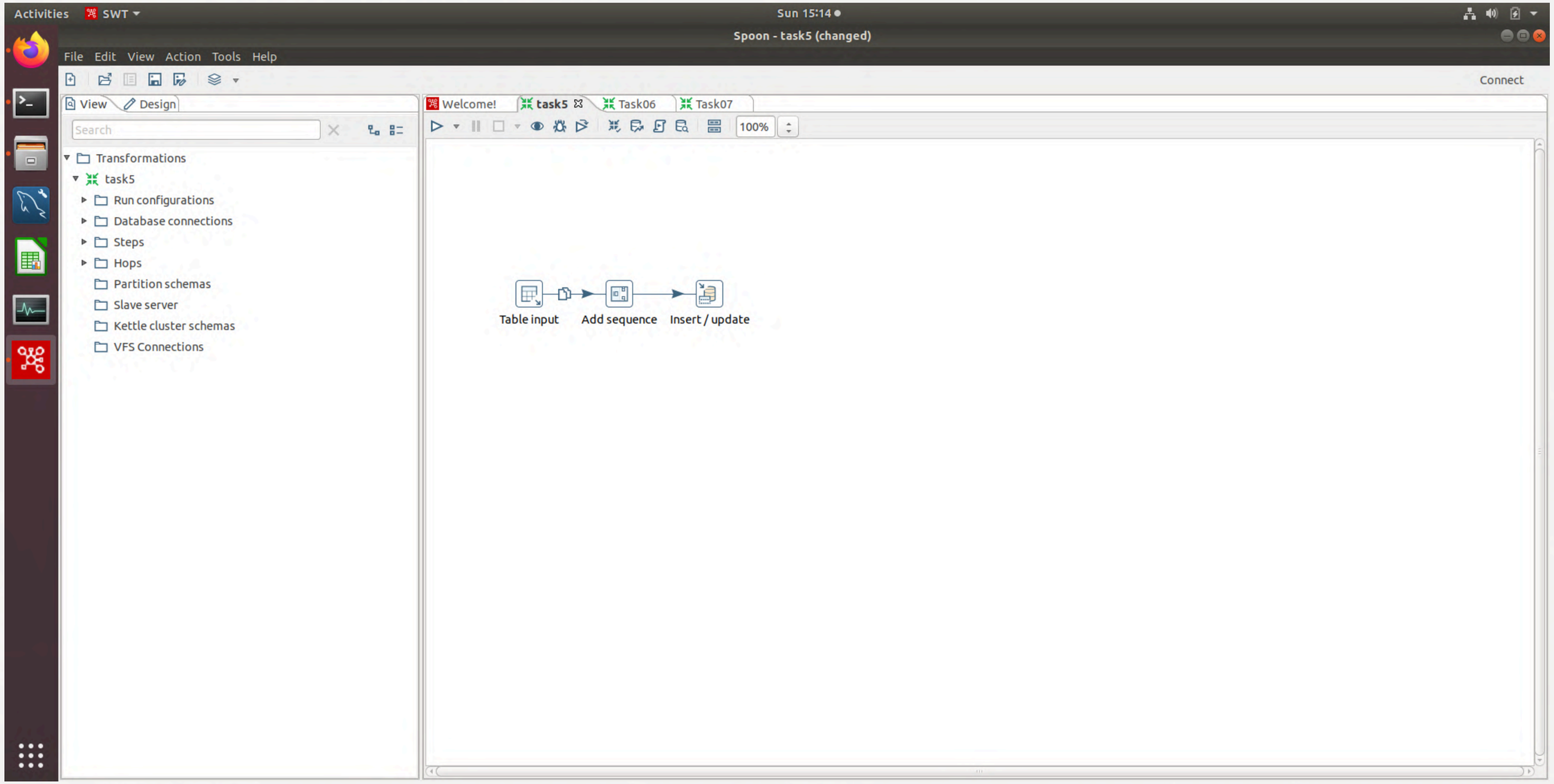
USE pp_star_database;

CREATE TABLE IF NOT EXISTS dim_location (
    locationid   INT,
    postcode     VARCHAR(10)    ,
    paon         VARCHAR(100)   ,
    saon         VARCHAR(50)    ,
    streetname   VARCHAR(50)    ,
    locality     VARCHAR(50)    ,
    city         VARCHAR(50)    ,
    district     VARCHAR(50)    ,
    county       VARCHAR(50)    ,
    PRIMARY KEY (locationid)
);

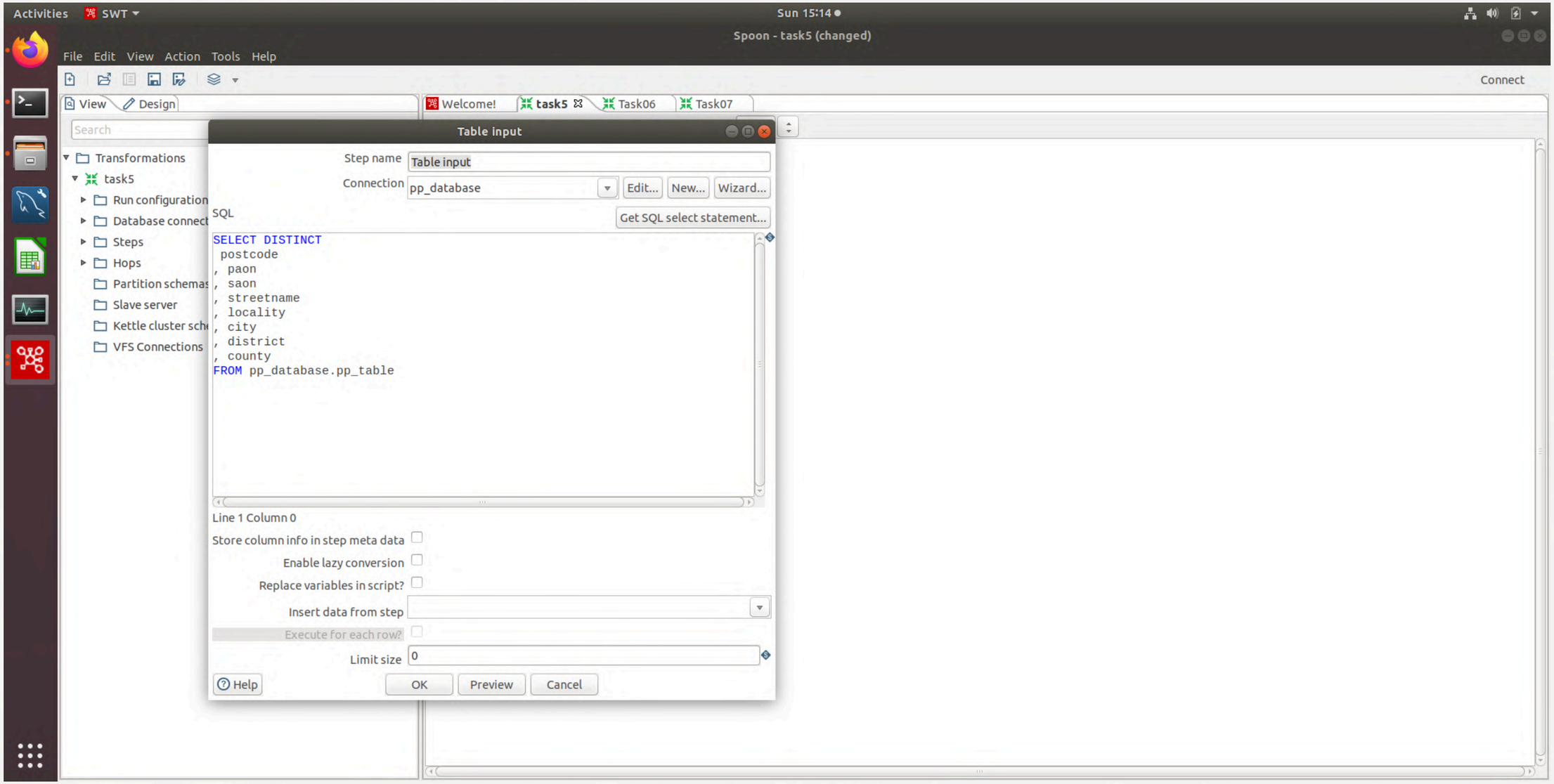
CREATE TABLE IF NOT EXISTS dim_time (
    timeid     DATETIME,
    day        INT,
    month_id   INT,
    month_name VARCHAR(50),
    year_id    INT,
    PRIMARY KEY (timeid)
);

CREATE TABLE IF NOT EXISTS fact_transaction (
    locationid INT,
    timeid     DATETIME,
    price      BIGINT,
    PRIMARY KEY (locationid, timeid),
    FOREIGN KEY (timeid) REFERENCES dim_time (timeid),
    FOREIGN KEY (locationid) REFERENCES dim_location(locationid)
);
```

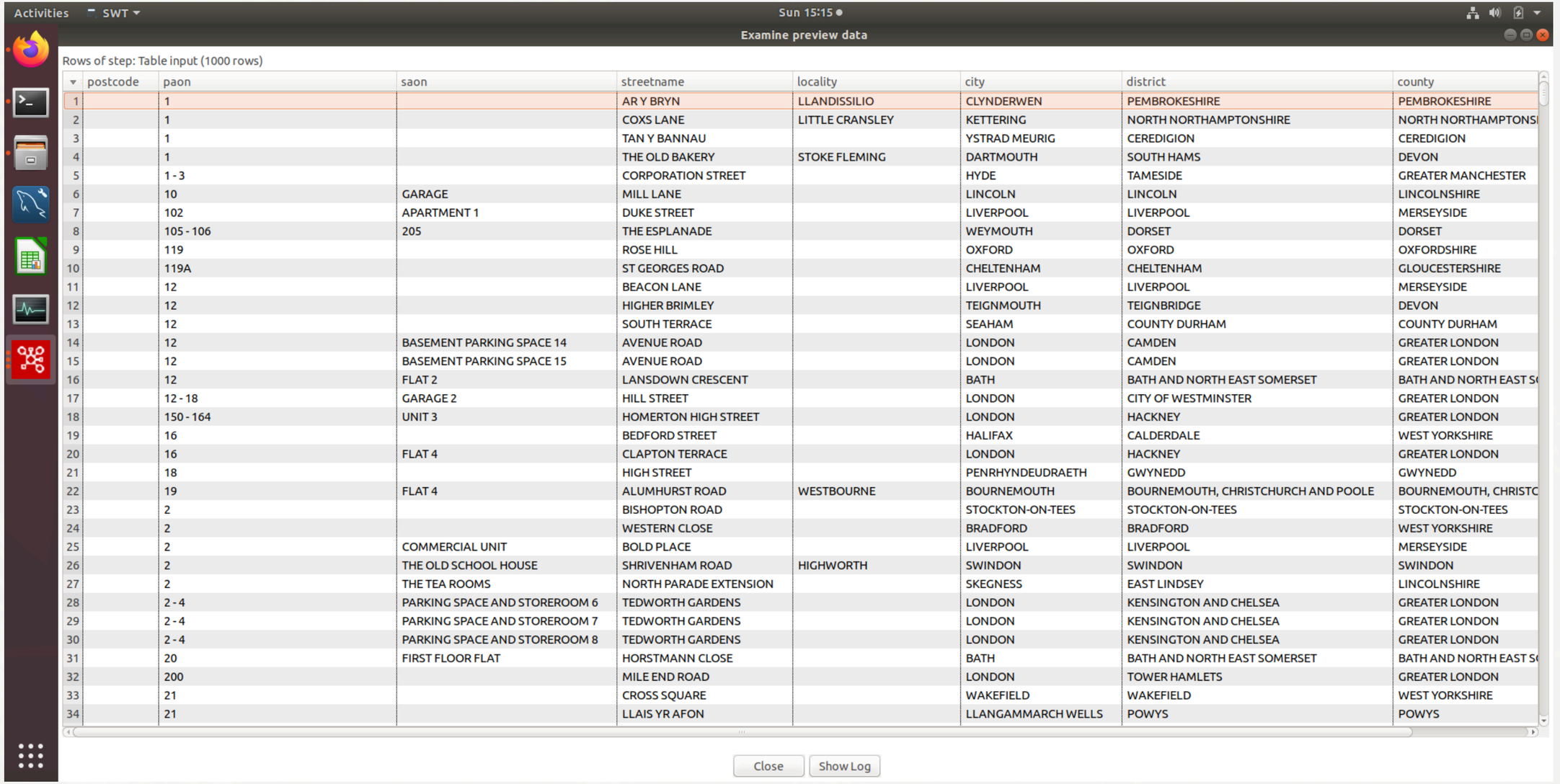
TASK 5 - Location Dimension



TASK 5 - Table Input configuration

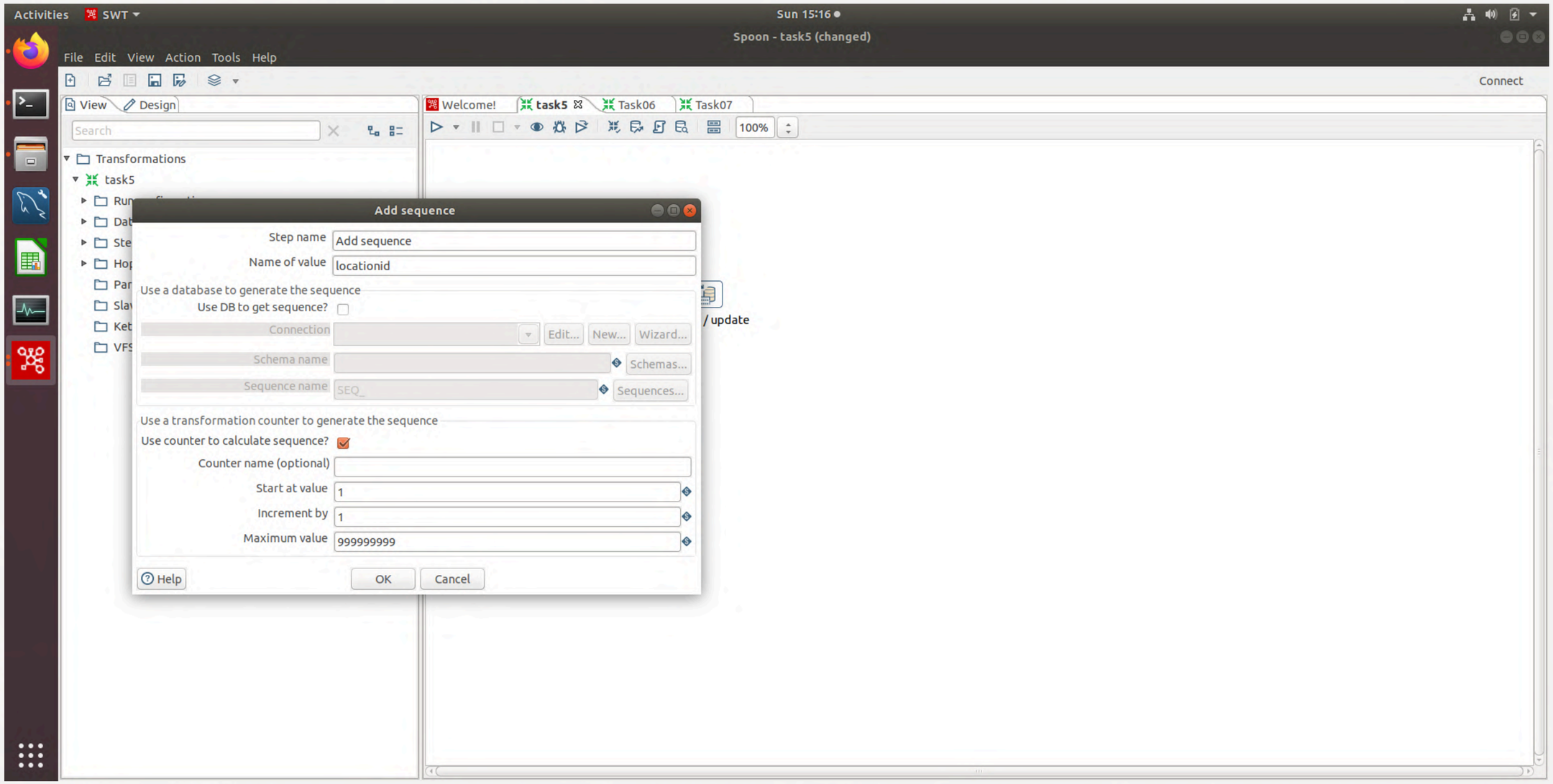


TASK 5 - Table Input preview

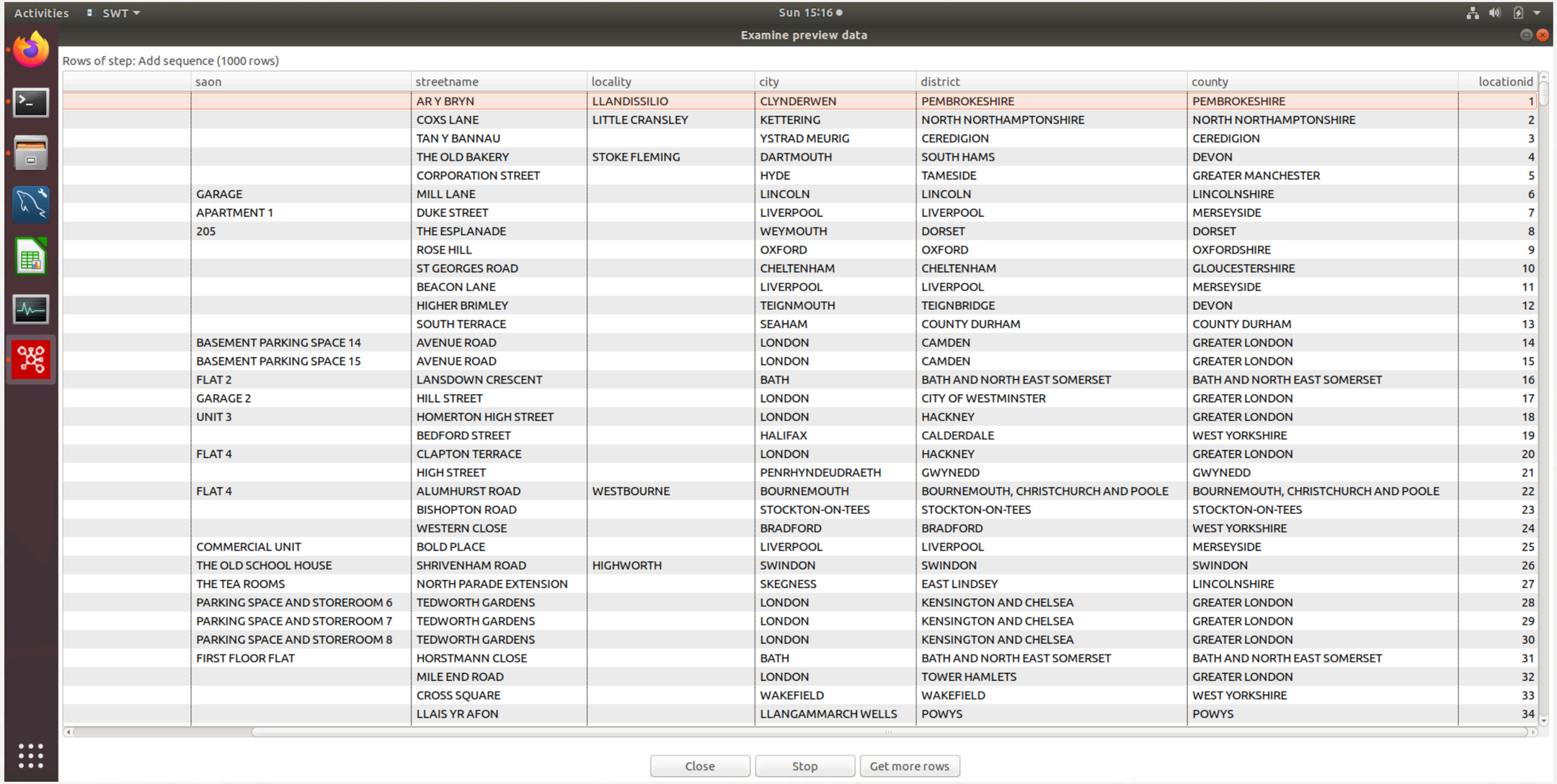


	postcode	paon	saon	streetname	locality	city	district	county
1		1		AR Y BRYN	LLANDISSILIO	CLYNDERWEN	PEMBROKESHIRE	PEMBROKESHIRE
2		1		COXS LANE	LITTLE CRANSLEY	KETTERING	NORTH NORTHAMPTONSHIRE	NORTH NORTHAMPTONSI
3		1		TAN Y BANNAU		YSTRAD MEURIG	CEREDIGION	CEREDIGION
4		1		THE OLD BAKERY	STOKE FLEMING	DARTMOUTH	SOUTH HAMS	DEVON
5		1 - 3		CORPORATION STREET		HYDE	TAMESIDE	GREATER MANCHESTER
6		10	GARAGE	MILL LANE		LINCOLN	LINCOLN	LINCOLNSHIRE
7		102	APARTMENT 1	DUKE STREET		LIVERPOOL	LIVERPOOL	MERSEYSIDE
8		105 - 106	205	THE ESPLANADE		WEYMOUTH	DORSET	DORSET
9		119		ROSE HILL		OXFORD	OXFORD	OXFORDSHIRE
10		119A		ST GEORGES ROAD		CHELTENHAM	CHELTENHAM	GLOUCESTERSHIRE
11		12		BEACON LANE		LIVERPOOL	LIVERPOOL	MERSEYSIDE
12		12		HIGHER BRIMLEY		TEIGNMOUTH	TEIGNBRIDGE	DEVON
13		12		SOUTH TERRACE		SEAHAM	COUNTY DURHAM	COUNTY DURHAM
14		12	BASEMENT PARKING SPACE 14	AVENUE ROAD		LONDON	CAMDEN	GREATER LONDON
15		12	BASEMENT PARKING SPACE 15	AVENUE ROAD		LONDON	CAMDEN	GREATER LONDON
16		12	FLAT 2	LANSDOWN CRESCENT		BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST S
17		12 - 18	GARAGE 2	HILL STREET		LONDON	CITY OF WESTMINSTER	GREATER LONDON
18		150 - 164	UNIT 3	HOMERTON HIGH STREET		LONDON	HACKNEY	GREATER LONDON
19		16		BEDFORD STREET		HALIFAX	CALDERDALE	WEST YORKSHIRE
20		16	FLAT 4	CLAPTON TERRACE		LONDON	HACKNEY	GREATER LONDON
21		18		HIGH STREET		PENRHYNDEUDRAETH	GWYNEDD	GWYNEDD
22		19	FLAT 4	ALUMHURST ROAD	WESTBOURNE	BOURNEMOUTH	BOURNEMOUTH, CHRISTCHURCH AND POOLE	BOURNEMOUTH, CHRISTC
23		2		BISHOPTON ROAD		STOCKTON-ON-TEES	STOCKTON-ON-TEES	STOCKTON-ON-TEES
24		2		WESTERN CLOSE		BRADFORD	BRADFORD	WEST YORKSHIRE
25		2	COMMERCIAL UNIT	BOLD PLACE		LIVERPOOL	LIVERPOOL	MERSEYSIDE
26		2	THE OLD SCHOOL HOUSE	SHRIVENHAM ROAD	HIGHWORTH	SWINDON	SWINDON	SWINDON
27		2	THE TEA ROOMS	NORTH PARADE EXTENSION		SKEGNESS	EAST LINDSEY	LINCOLNSHIRE
28		2 - 4	PARKING SPACE AND STOREROOM 6	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON
29		2 - 4	PARKING SPACE AND STOREROOM 7	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON
30		2 - 4	PARKING SPACE AND STOREROOM 8	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON
31		20	FIRST FLOOR FLAT	HORSTMANN CLOSE		BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST S
32		200		MILE END ROAD		LONDON	TOWER HAMLETS	GREATER LONDON
33		21		CROSS SQUARE		WAKEFIELD	WAKEFIELD	WEST YORKSHIRE
34		21		LLAIS YR AFON		LLANGAMMARCH WELLS	POWYS	POWYS

TASK 5 - Add Sequence configuration

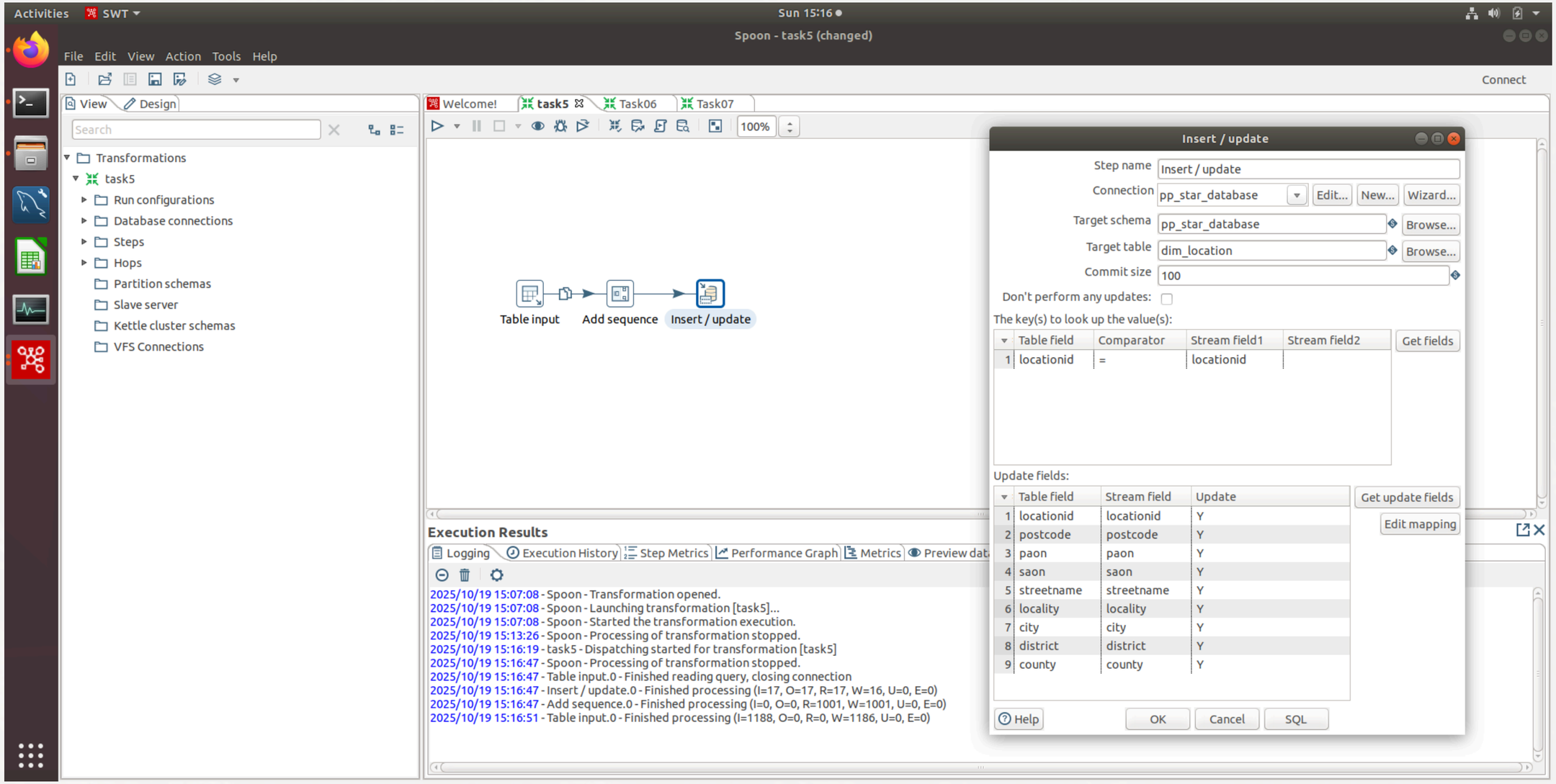


TASK 5 - Add Sequence preview

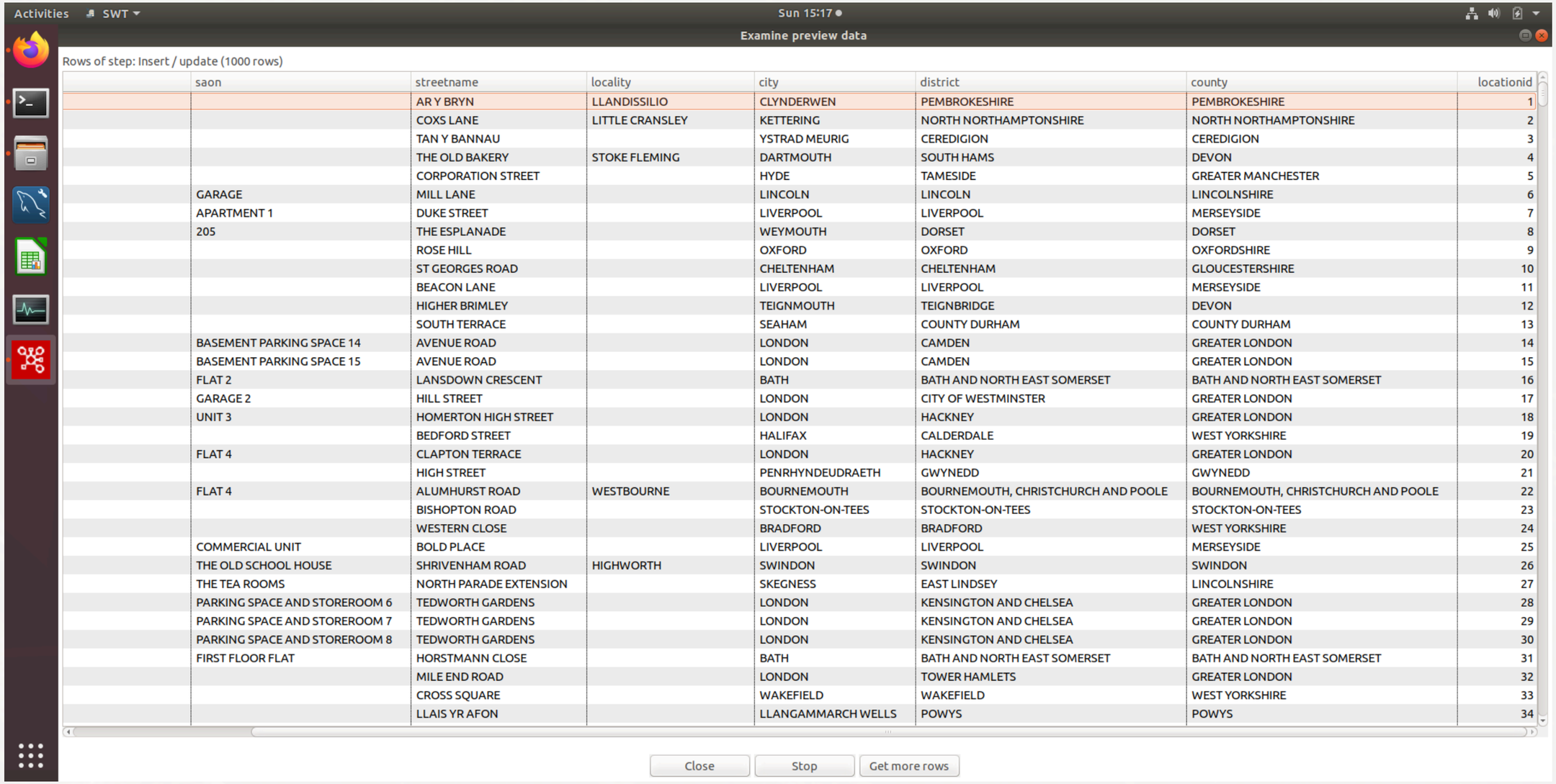


	saon	streetname	locality	city	district	county	locationid
		AR Y BRYN	LLANDISSILIO	CLYNDERWEN	PEMBROKESHIRE	PEMBROKESHIRE	1
		COXS LANE	LITTLE CRANSLEY	KETTERING	NORTH NORTHAMPTONSHIRE	NORTH NORTHAMPTONSHIRE	2
		TAN Y BANNAU		YSTRAD MEURIG	CEREDIGION	CEREDIGION	3
		THE OLD BAKERY	STOKE FLEMING	DARTMOUTH	SOUTH HAMS	DEVON	4
		CORPORATION STREET		HYDE	TAMESIDE	GREATER MANCHESTER	5
	GARAGE	MILL LANE		LINCOLN	LINCOLN	LINCOLNSHIRE	6
	APARTMENT 1	DUKE STREET		LIVERPOOL	LIVERPOOL	MERSEYSIDE	7
	205	THE ESPLANADE		WEYMOUTH	DORSET	DORSET	8
		ROSE HILL		OXFORD	OXFORD	OXFORDSHIRE	9
		ST GEORGES ROAD		CHELTENHAM	CHELTENHAM	GLOUCESTERSHIRE	10
		BEACON LANE		LIVERPOOL	LIVERPOOL	MERSEYSIDE	11
		HIGHER BRIMLEY		TEIGNMOUTH	TEIGNBRIDGE	DEVON	12
		SOUTH TERRACE		SEAHAM	COUNTY DURHAM	COUNTY DURHAM	13
	BASEMENT PARKING SPACE 14	AVENUE ROAD		LONDON	CAMDEN	GREATER LONDON	14
	BASEMENT PARKING SPACE 15	AVENUE ROAD		LONDON	CAMDEN	GREATER LONDON	15
	FLAT 2	LANSDOWN CRESCENT		BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST SOMERSET	16
	GARAGE 2	HILL STREET		LONDON	CITY OF WESTMINSTER	GREATER LONDON	17
	UNIT 3	HOMERTON HIGH STREET		LONDON	HACKNEY	GREATER LONDON	18
		BEDFORD STREET		HALIFAX	CALDERDALE	WEST YORKSHIRE	19
	FLAT 4	CLAPTON TERRACE		LONDON	HACKNEY	GREATER LONDON	20
		HIGH STREET		PENRHYNDEUDRAETH	GWYNEDD	GWYNEDD	21
	FLAT 4	ALUMHURST ROAD	WESTBOURNE	BOURNEMOUTH	BOURNEMOUTH, CHRISTCHURCH AND POOLE	BOURNEMOUTH, CHRISTCHURCH AND POOLE	22
		BISHOPTON ROAD		STOCKTON-ON-TEES	STOCKTON-ON-TEES	STOCKTON-ON-TEES	23
		WESTERN CLOSE		BRADFORD	BRADFORD	WEST YORKSHIRE	24
	COMMERCIAL UNIT	BOLD PLACE		LIVERPOOL	LIVERPOOL	MERSEYSIDE	25
	THE OLD SCHOOL HOUSE	SHRIVENHAM ROAD	HIGHWORTH	SWINDON	SWINDON	SWINDON	26
	THE TEA ROOMS	NORTH PARADE EXTENSION		SKEGNESS	EAST LINDSEY	LINCOLNSHIRE	27
	PARKING SPACE AND STOREROOM 6	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	28
	PARKING SPACE AND STOREROOM 7	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	29
	PARKING SPACE AND STOREROOM 8	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	30
	FIRST FLOOR FLAT	HORSTMANN CLOSE		BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST SOMERSET	31
		MILE END ROAD		LONDON	TOWER HAMLETS	GREATER LONDON	32
		CROSS SQUARE		WAKEFIELD	WAKEFIELD	WEST YORKSHIRE	33
		LLAIS YR AFON		LLANGAMMARCH WELLS	POWYS	POWYS	34

TASK 5 - Insert/Update configuration

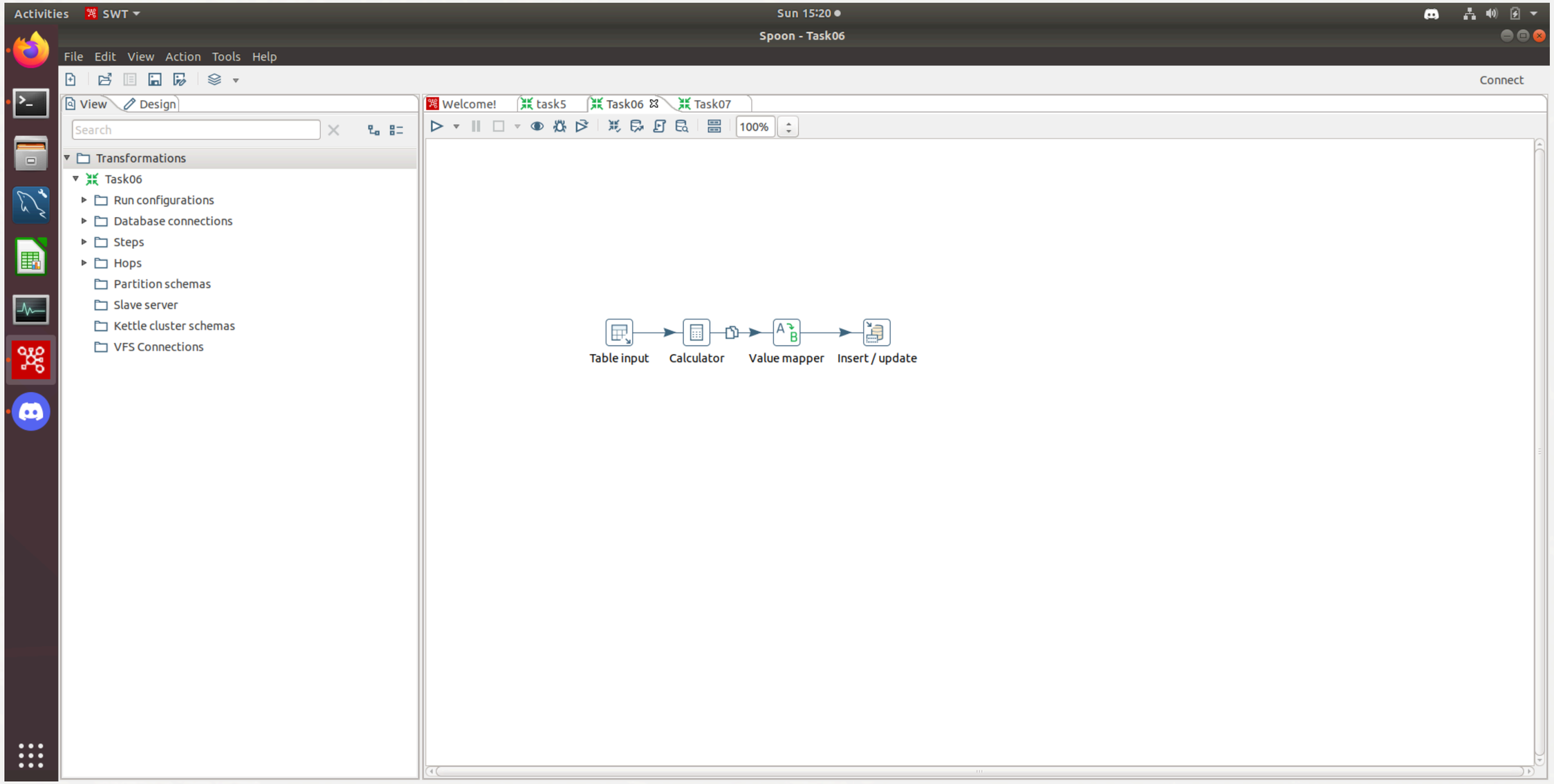


TASK 5 - Insert/Update configuration

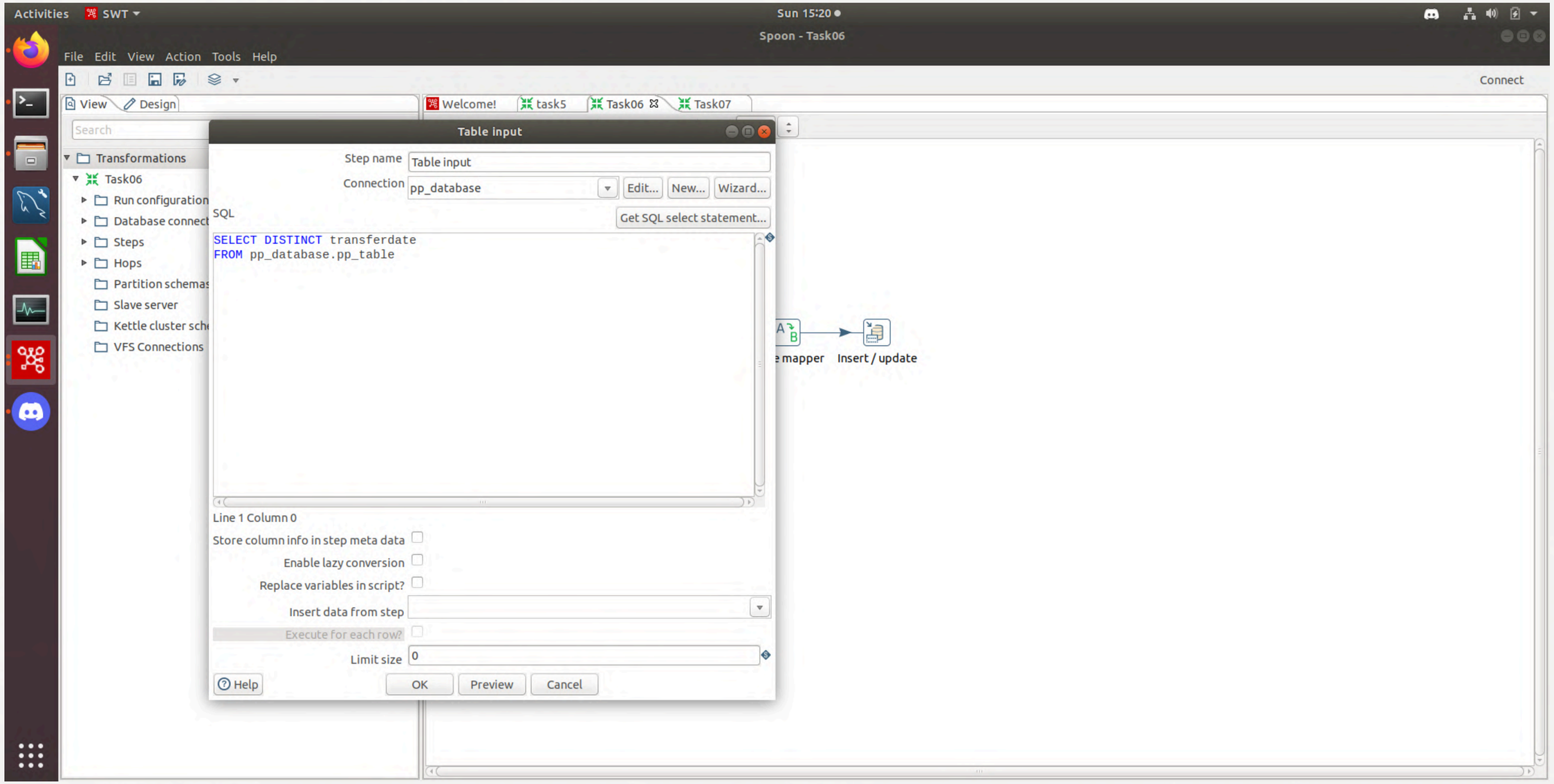


	saon	streetname	locality	city	district	county	locationid
		AR Y BRYN	LLANDISSILIO	CLYNDERWEN	PEMBROKESHIRE	PEMBROKESHIRE	1
		COXS LANE	LITTLE CRANSLEY	KETTERING	NORTH NORTHAMPTONSHIRE	NORTH NORTHAMPTONSHIRE	2
		TAN Y BANNAU		YSTRAD MEURIG	CEREDIGION	CEREDIGION	3
		THE OLD BAKERY	STOKE FLEMING	DARTMOUTH	SOUTH HAMS	DEVON	4
		CORPORATION STREET		HYDE	TAMESIDE	GREATER MANCHESTER	5
	GARAGE	MILL LANE		LINCOLN	LINCOLN	LINCOLNSHIRE	6
	APARTMENT 1	DUKE STREET		LIVERPOOL	LIVERPOOL	MERSEYSIDE	7
	205	THE ESPLANADE		WEYMOUTH	DORSET	DORSET	8
		ROSE HILL		OXFORD	OXFORD	OXFORDSHIRE	9
		ST GEORGES ROAD		CHELTENHAM	CHELTENHAM	GLOUCESTERSHIRE	10
		BEACON LANE		LIVERPOOL	LIVERPOOL	MERSEYSIDE	11
		HIGHER BRIMLEY		TEIGNMOUTH	TEIGNBRIDGE	DEVON	12
		SOUTH TERRACE		SEAHAM	COUNTY DURHAM	COUNTY DURHAM	13
	BASEMENT PARKING SPACE 14	AVENUE ROAD		LONDON	CAMDEN	GREATER LONDON	14
	BASEMENT PARKING SPACE 15	AVENUE ROAD		LONDON	CAMDEN	GREATER LONDON	15
	FLAT 2	LANSDOWN CRESCENT		BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST SOMERSET	16
	GARAGE 2	HILL STREET		LONDON	CITY OF WESTMINSTER	GREATER LONDON	17
	UNIT 3	HOMERTON HIGH STREET		LONDON	HACKNEY	GREATER LONDON	18
		BEDFORD STREET		HALIFAX	CALDERDALE	WEST YORKSHIRE	19
	FLAT 4	CLAPTON TERRACE		LONDON	HACKNEY	GREATER LONDON	20
		HIGH STREET		PENRHYNDEUDRAETH	GWYNEDD	GWYNEDD	21
	FLAT 4	ALUMHURST ROAD	WESTBOURNE	BOURNEMOUTH	BOURNEMOUTH, CHRISTCHURCH AND POOLE	BOURNEMOUTH, CHRISTCHURCH AND POOLE	22
		BISHOPTON ROAD		STOCKTON-ON-TEES	STOCKTON-ON-TEES	STOCKTON-ON-TEES	23
		WESTERN CLOSE		BRADFORD	BRADFORD	WEST YORKSHIRE	24
	COMMERCIAL UNIT	BOLD PLACE		LIVERPOOL	LIVERPOOL	MERSEYSIDE	25
	THE OLD SCHOOL HOUSE	SHRIVENHAM ROAD	HIGHWORTH	SWINDON	SWINDON	SWINDON	26
	THE TEA ROOMS	NORTH PARADE EXTENSION		SKEGNESS	EAST LINDSEY	LINCOLNSHIRE	27
	PARKING SPACE AND STOREROOM 6	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	28
	PARKING SPACE AND STOREROOM 7	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	29
	PARKING SPACE AND STOREROOM 8	TEDWORTH GARDENS		LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	30
	FIRST FLOOR FLAT	HORSTMANN CLOSE		BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST SOMERSET	31
		MILE END ROAD		LONDON	TOWER HAMLETS	GREATER LONDON	32
		CROSS SQUARE		WAKEFIELD	WAKEFIELD	WEST YORKSHIRE	33
		LLAIS YR AFON		LLANGAMMARCH WELLS	POWYS	POWYS	34

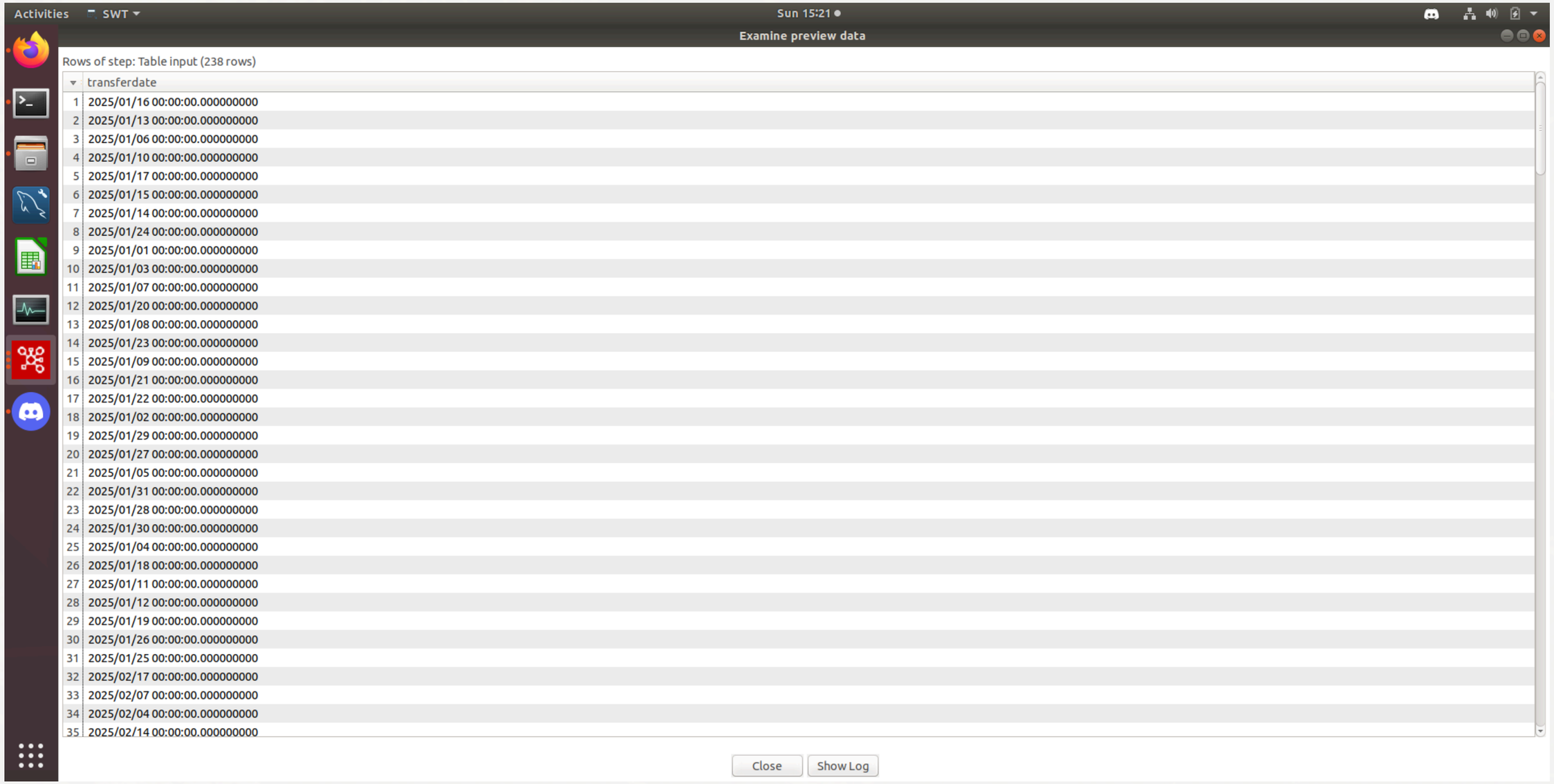
TASK 6 - Time Dimension



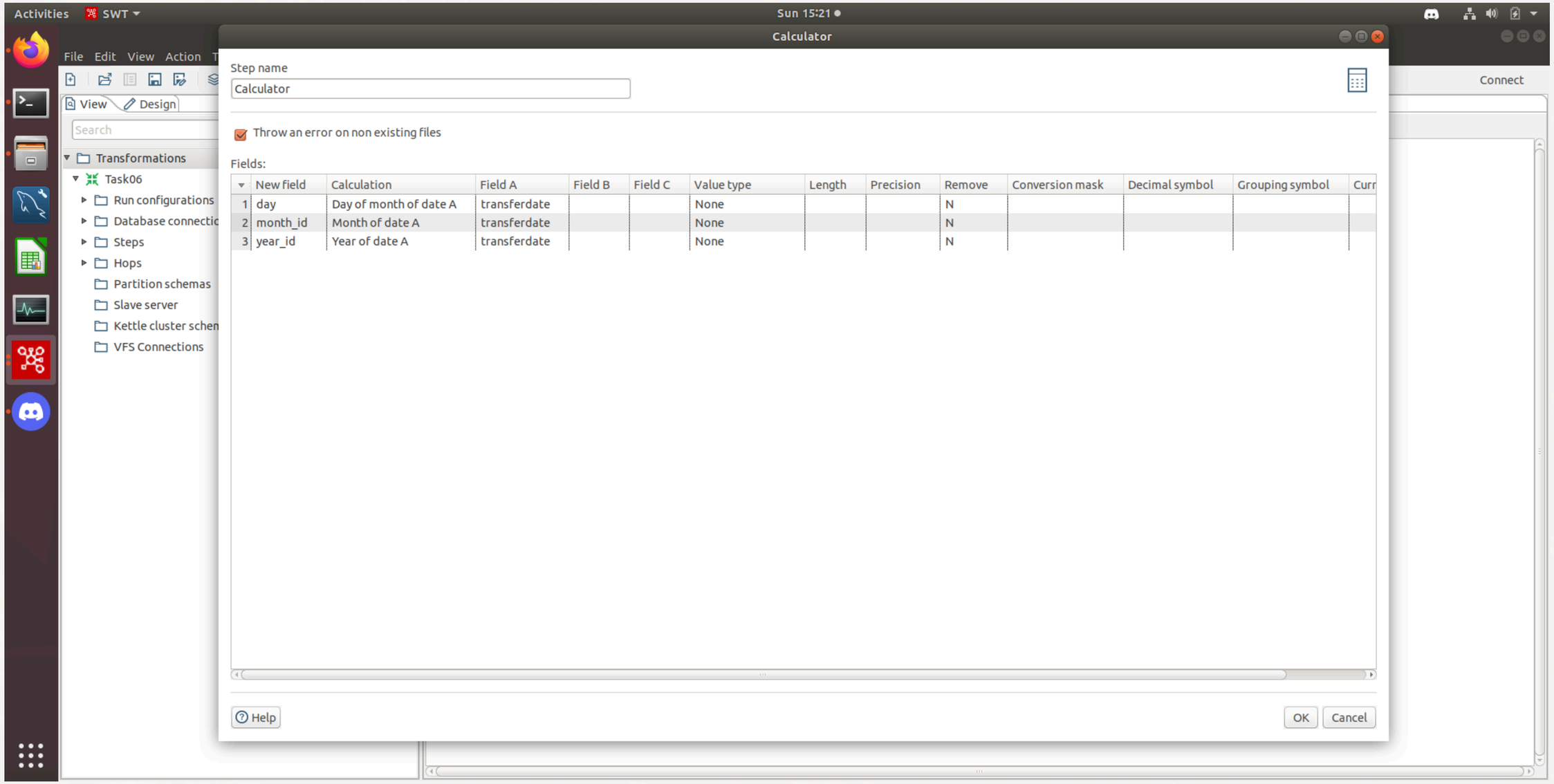
TASK 6 - Table Input configuration



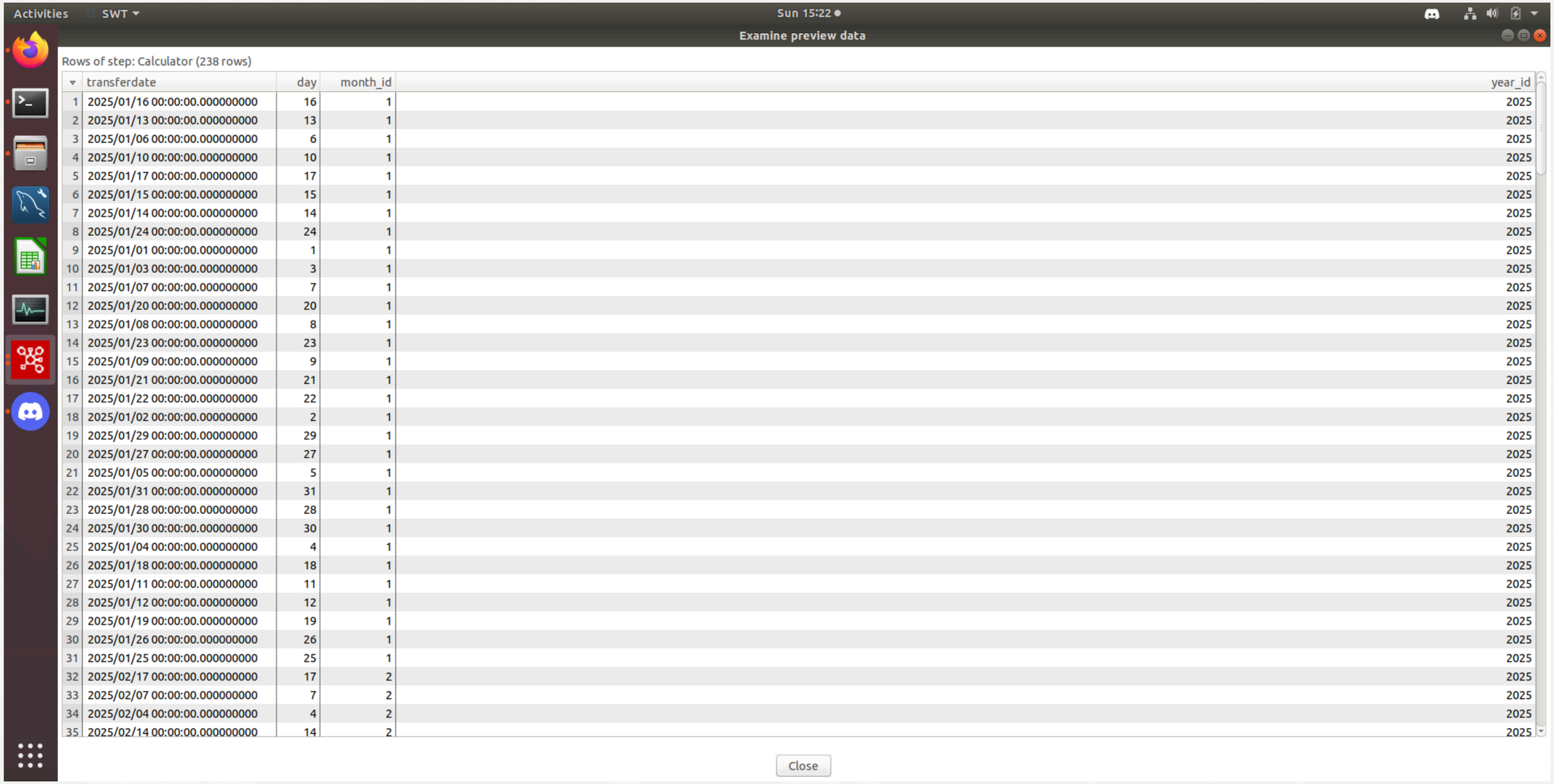
TASK 6 - Table Input preview



TASK 6 - Calculator configuration

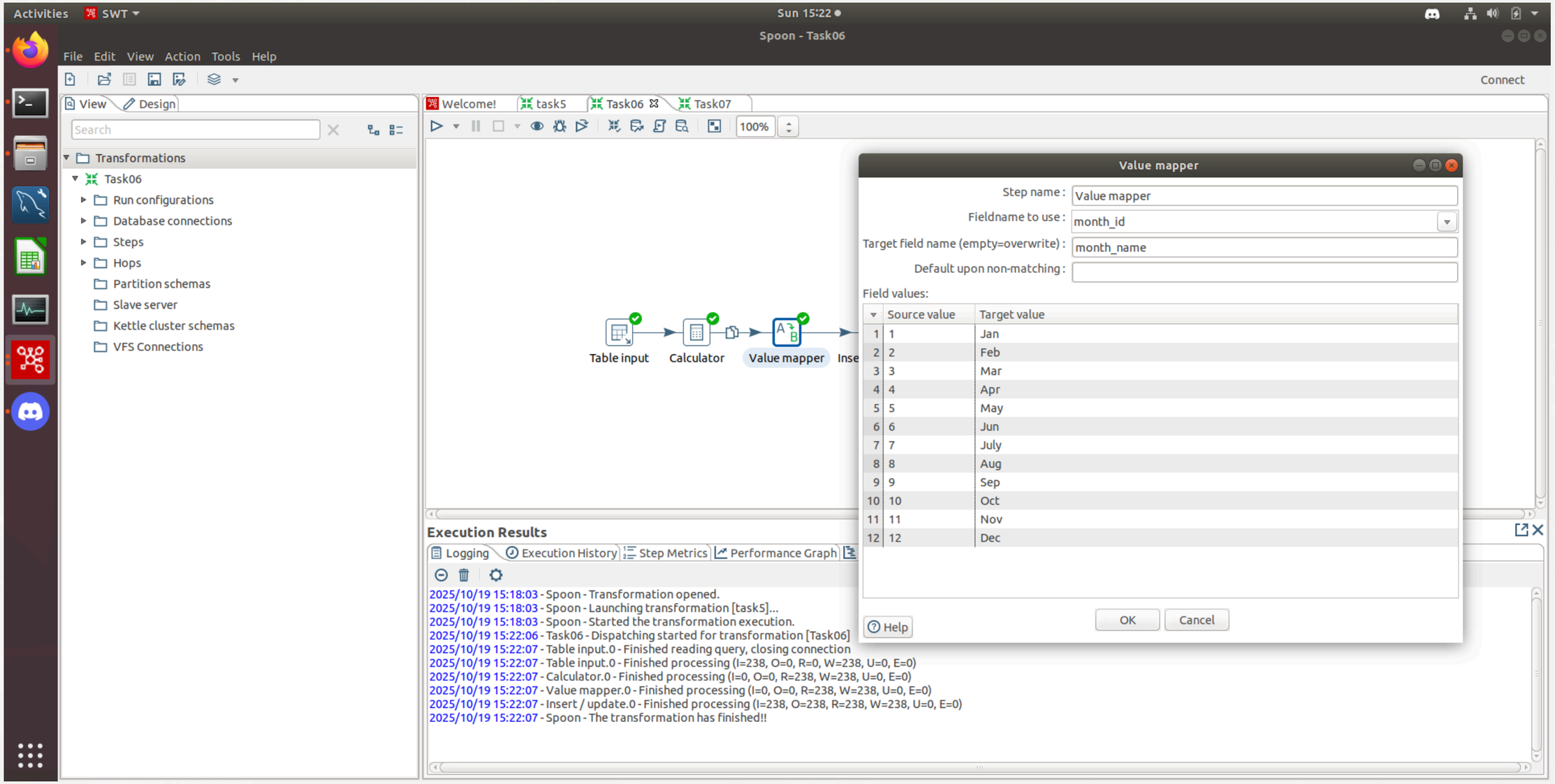


TASK 6 - Calculator preview

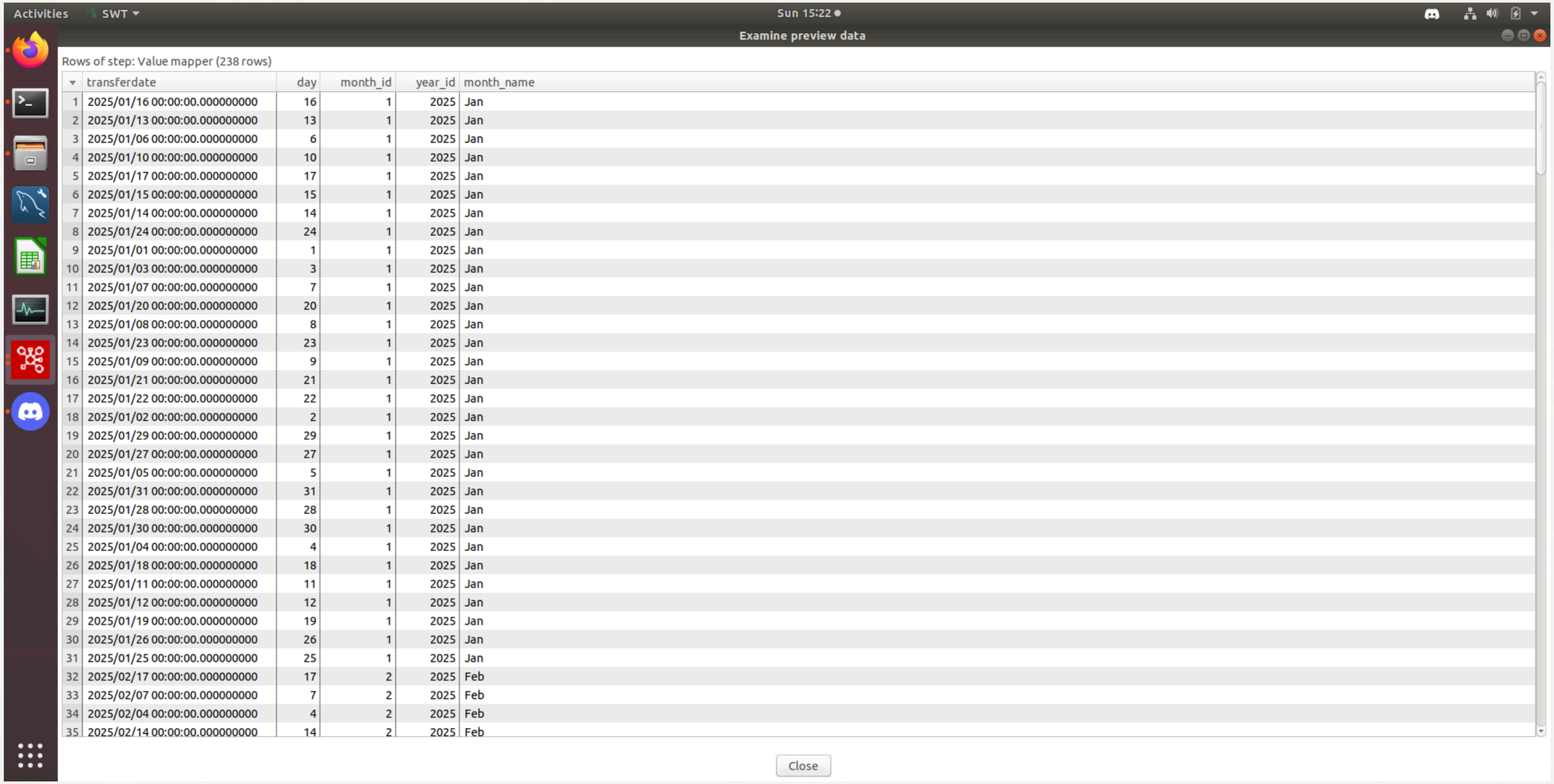


	transferdate	day	month_id	year_id
1	2025/01/16 00:00:00.000000000	16	1	2025
2	2025/01/13 00:00:00.000000000	13	1	2025
3	2025/01/06 00:00:00.000000000	6	1	2025
4	2025/01/10 00:00:00.000000000	10	1	2025
5	2025/01/17 00:00:00.000000000	17	1	2025
6	2025/01/15 00:00:00.000000000	15	1	2025
7	2025/01/14 00:00:00.000000000	14	1	2025
8	2025/01/24 00:00:00.000000000	24	1	2025
9	2025/01/01 00:00:00.000000000	1	1	2025
10	2025/01/03 00:00:00.000000000	3	1	2025
11	2025/01/07 00:00:00.000000000	7	1	2025
12	2025/01/20 00:00:00.000000000	20	1	2025
13	2025/01/08 00:00:00.000000000	8	1	2025
14	2025/01/23 00:00:00.000000000	23	1	2025
15	2025/01/09 00:00:00.000000000	9	1	2025
16	2025/01/21 00:00:00.000000000	21	1	2025
17	2025/01/22 00:00:00.000000000	22	1	2025
18	2025/01/02 00:00:00.000000000	2	1	2025
19	2025/01/29 00:00:00.000000000	29	1	2025
20	2025/01/27 00:00:00.000000000	27	1	2025
21	2025/01/05 00:00:00.000000000	5	1	2025
22	2025/01/31 00:00:00.000000000	31	1	2025
23	2025/01/28 00:00:00.000000000	28	1	2025
24	2025/01/30 00:00:00.000000000	30	1	2025
25	2025/01/04 00:00:00.000000000	4	1	2025
26	2025/01/18 00:00:00.000000000	18	1	2025
27	2025/01/11 00:00:00.000000000	11	1	2025
28	2025/01/12 00:00:00.000000000	12	1	2025
29	2025/01/19 00:00:00.000000000	19	1	2025
30	2025/01/26 00:00:00.000000000	26	1	2025
31	2025/01/25 00:00:00.000000000	25	1	2025
32	2025/02/17 00:00:00.000000000	17	2	2025
33	2025/02/07 00:00:00.000000000	7	2	2025
34	2025/02/04 00:00:00.000000000	4	2	2025
35	2025/02/14 00:00:00.000000000	14	2	2025

TASK 6 - Value Mapper configuration

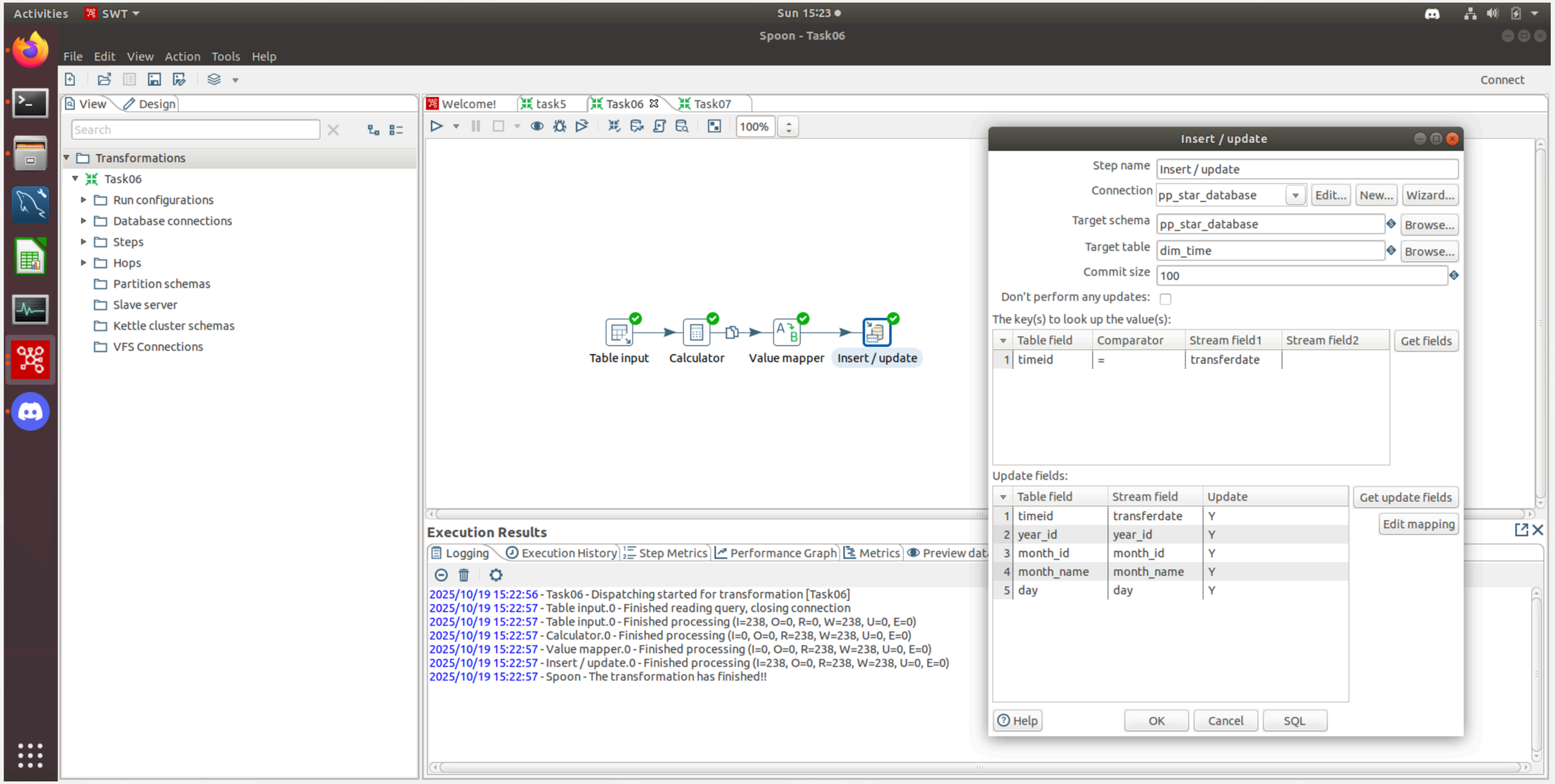


TASK 6 - Value Mapper preview

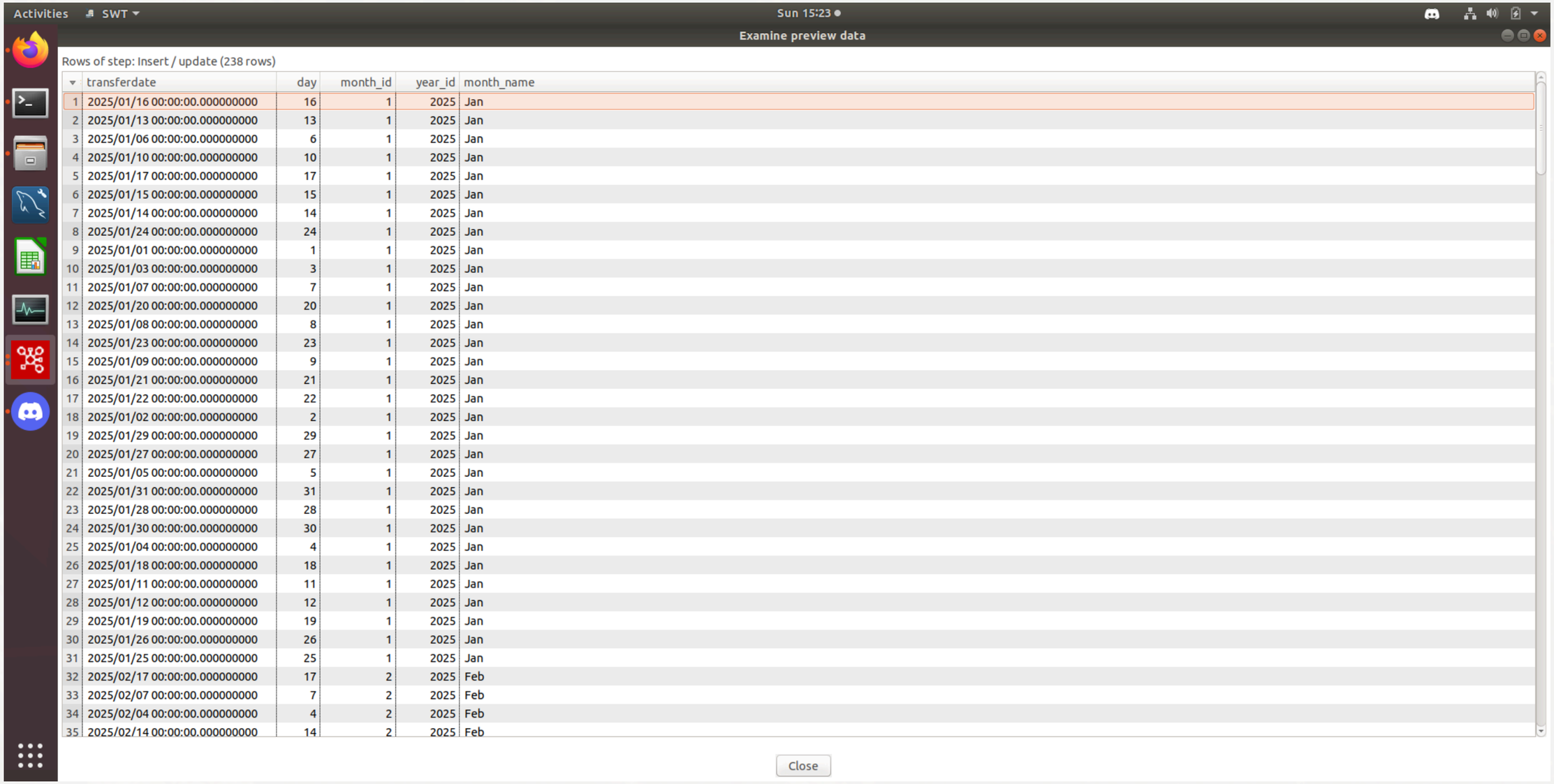


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3	2025/01/06 00:00:00.000000000	6	1	2025	Jan
4	2025/01/10 00:00:00.000000000	10	1	2025	Jan
5	2025/01/17 00:00:00.000000000	17	1	2025	Jan
6	2025/01/15 00:00:00.000000000	15	1	2025	Jan
7	2025/01/14 00:00:00.000000000	14	1	2025	Jan
8	2025/01/24 00:00:00.000000000	24	1	2025	Jan
9	2025/01/01 00:00:00.000000000	1	1	2025	Jan
10	2025/01/03 00:00:00.000000000	3	1	2025	Jan
11	2025/01/07 00:00:00.000000000	7	1	2025	Jan
12	2025/01/20 00:00:00.000000000	20	1	2025	Jan
13	2025/01/08 00:00:00.000000000	8	1	2025	Jan
14	2025/01/23 00:00:00.000000000	23	1	2025	Jan
15	2025/01/09 00:00:00.000000000	9	1	2025	Jan
16	2025/01/21 00:00:00.000000000	21	1	2025	Jan
17	2025/01/22 00:00:00.000000000	22	1	2025	Jan
18	2025/01/02 00:00:00.000000000	2	1	2025	Jan
19	2025/01/29 00:00:00.000000000	29	1	2025	Jan
20	2025/01/27 00:00:00.000000000	27	1	2025	Jan
21	2025/01/05 00:00:00.000000000	5	1	2025	Jan
22	2025/01/31 00:00:00.000000000	31	1	2025	Jan
23	2025/01/28 00:00:00.000000000	28	1	2025	Jan
24	2025/01/30 00:00:00.000000000	30	1	2025	Jan
25	2025/01/04 00:00:00.000000000	4	1	2025	Jan
26	2025/01/18 00:00:00.000000000	18	1	2025	Jan
27	2025/01/11 00:00:00.000000000	11	1	2025	Jan
28	2025/01/12 00:00:00.000000000	12	1	2025	Jan
29	2025/01/19 00:00:00.000000000	19	1	2025	Jan
30	2025/01/26 00:00:00.000000000	26	1	2025	Jan
31	2025/01/25 00:00:00.000000000	25	1	2025	Jan
32	2025/02/17 00:00:00.000000000	17	2	2025	Feb
33	2025/02/07 00:00:00.000000000	7	2	2025	Feb
34	2025/02/04 00:00:00.000000000	4	2	2025	Feb
35	2025/02/14 00:00:00.000000000	14	2	2025	Feb

TASK 6 - Insert/Update configuration

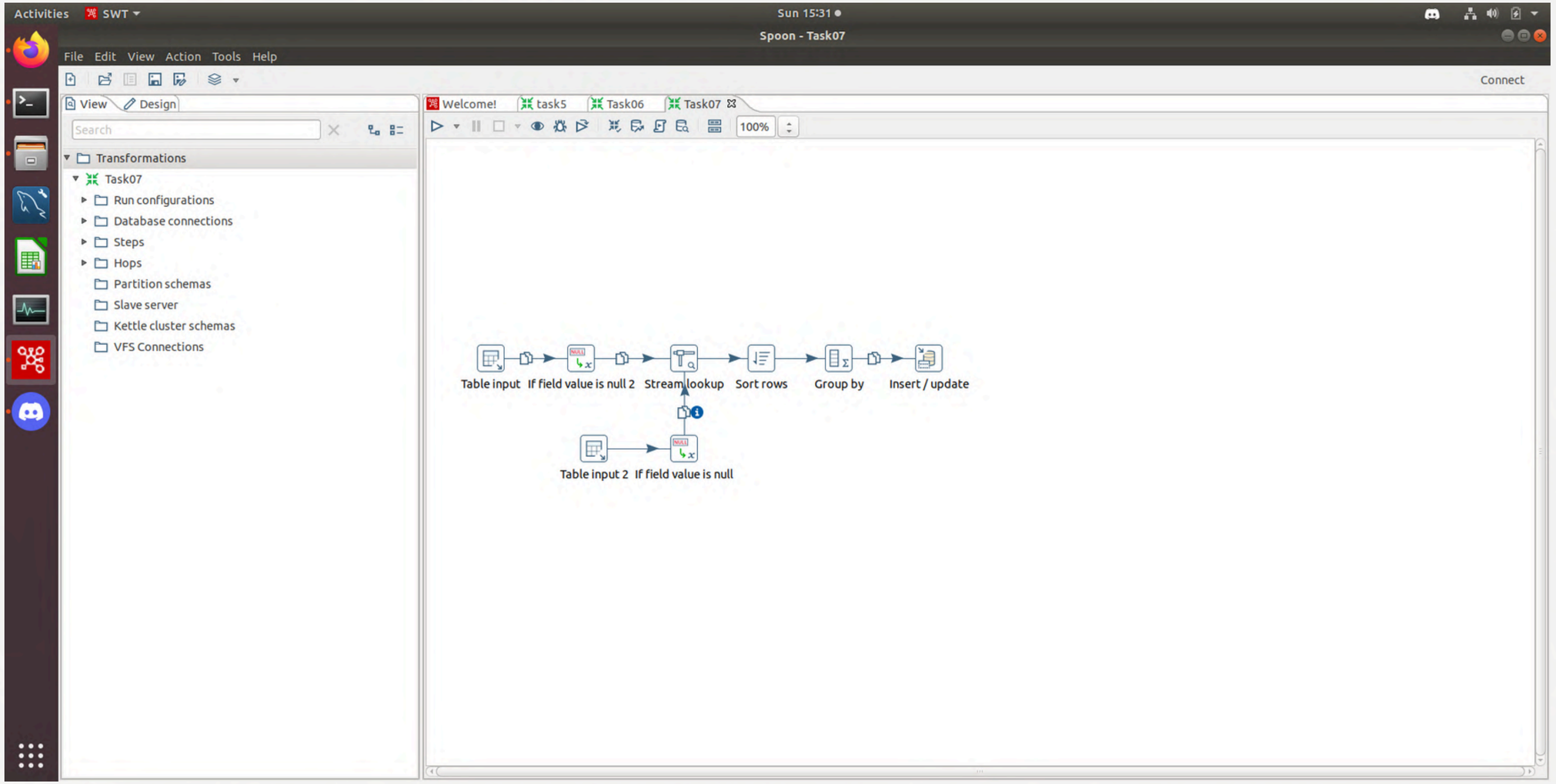


TASK 6 - Insert/Update preview



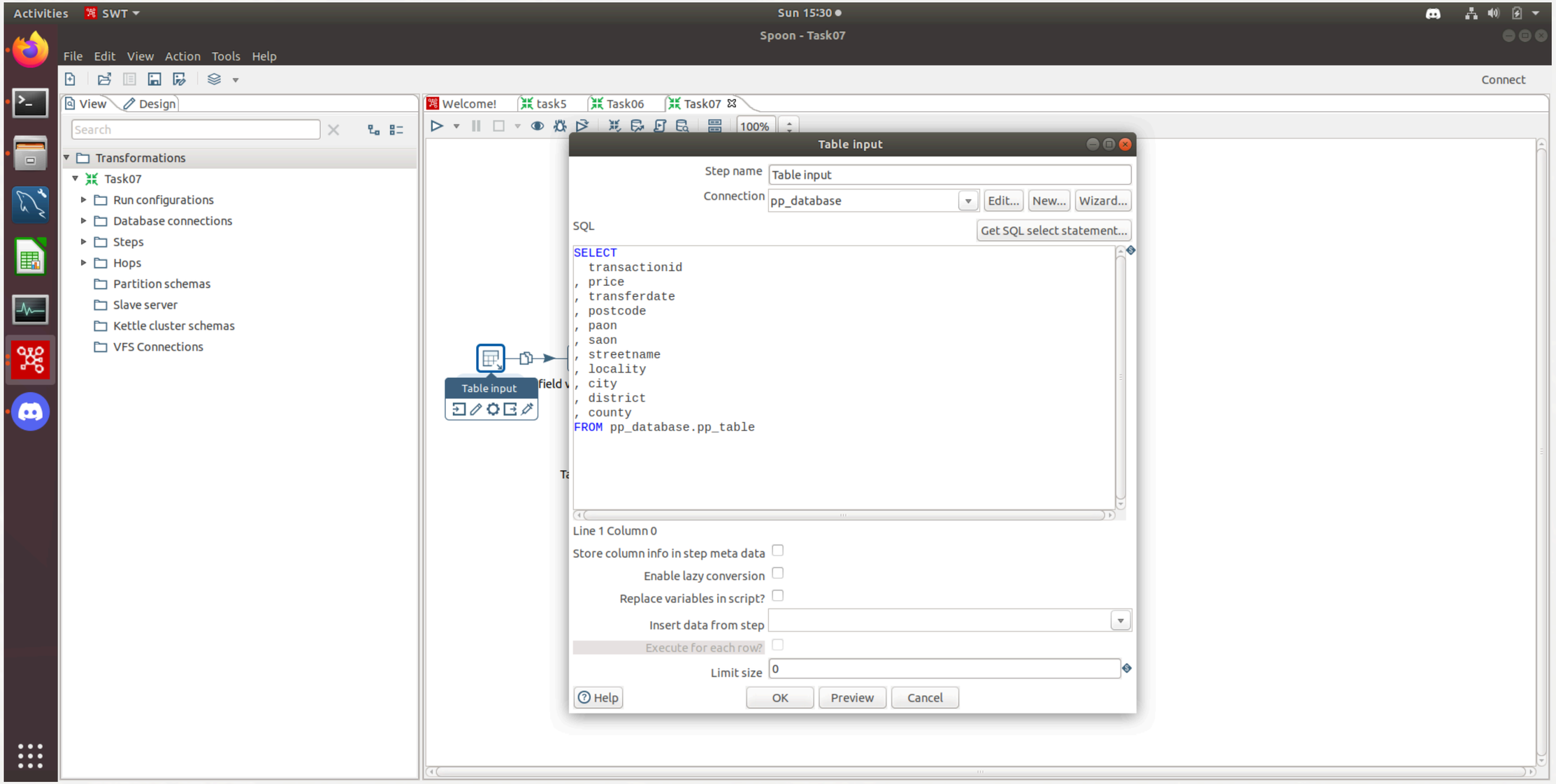
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4	2025/01/10 00:00:00.000000000	10	1	2025	Jan
5	2025/01/17 00:00:00.000000000	17	1	2025	Jan
6	2025/01/15 00:00:00.000000000	15	1	2025	Jan
7	2025/01/14 00:00:00.000000000	14	1	2025	Jan
8	2025/01/24 00:00:00.000000000	24	1	2025	Jan
9	2025/01/01 00:00:00.000000000	1	1	2025	Jan
10	2025/01/03 00:00:00.000000000	3	1	2025	Jan
11	2025/01/07 00:00:00.000000000	7	1	2025	Jan
12	2025/01/20 00:00:00.000000000	20	1	2025	Jan
13	2025/01/08 00:00:00.000000000	8	1	2025	Jan
14	2025/01/23 00:00:00.000000000	23	1	2025	Jan
15	2025/01/09 00:00:00.000000000	9	1	2025	Jan
16	2025/01/21 00:00:00.000000000	21	1	2025	Jan
17	2025/01/22 00:00:00.000000000	22	1	2025	Jan
18	2025/01/02 00:00:00.000000000	2	1	2025	Jan
19	2025/01/29 00:00:00.000000000	29	1	2025	Jan
20	2025/01/27 00:00:00.000000000	27	1	2025	Jan
21	2025/01/05 00:00:00.000000000	5	1	2025	Jan
22	2025/01/31 00:00:00.000000000	31	1	2025	Jan
23	2025/01/28 00:00:00.000000000	28	1	2025	Jan
24	2025/01/30 00:00:00.000000000	30	1	2025	Jan
25	2025/01/04 00:00:00.000000000	4	1	2025	Jan
26	2025/01/18 00:00:00.000000000	18	1	2025	Jan
27	2025/01/11 00:00:00.000000000	11	1	2025	Jan
28	2025/01/12 00:00:00.000000000	12	1	2025	Jan
29	2025/01/19 00:00:00.000000000	19	1	2025	Jan
30	2025/01/26 00:00:00.000000000	26	1	2025	Jan
31	2025/01/25 00:00:00.000000000	25	1	2025	Jan
32	2025/02/17 00:00:00.000000000	17	2	2025	Feb
33	2025/02/07 00:00:00.000000000	7	2	2025	Feb
34	2025/02/04 00:00:00.000000000	4	2	2025	Feb
35	2025/02/14 00:00:00.000000000	14	2	2025	Feb

TASK 7 - Fact Table

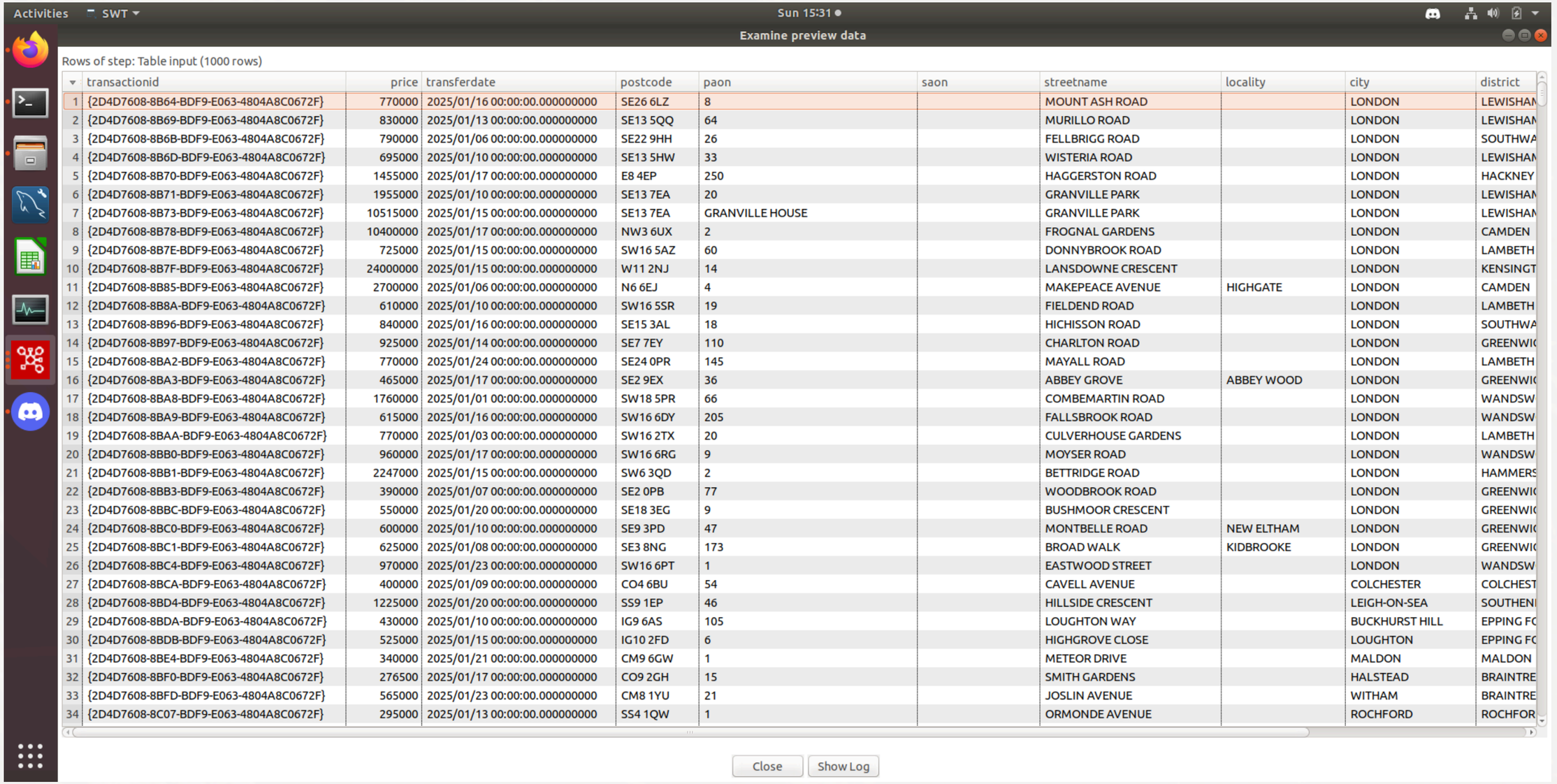


Note: Table Input (1) goes into “If field value is null 2”, and Table Input 2 goes into “If field value is null (1)”. Both “If field value is null” steps have the same configuration, however.

TASK 7 - Table Input (1) configuration

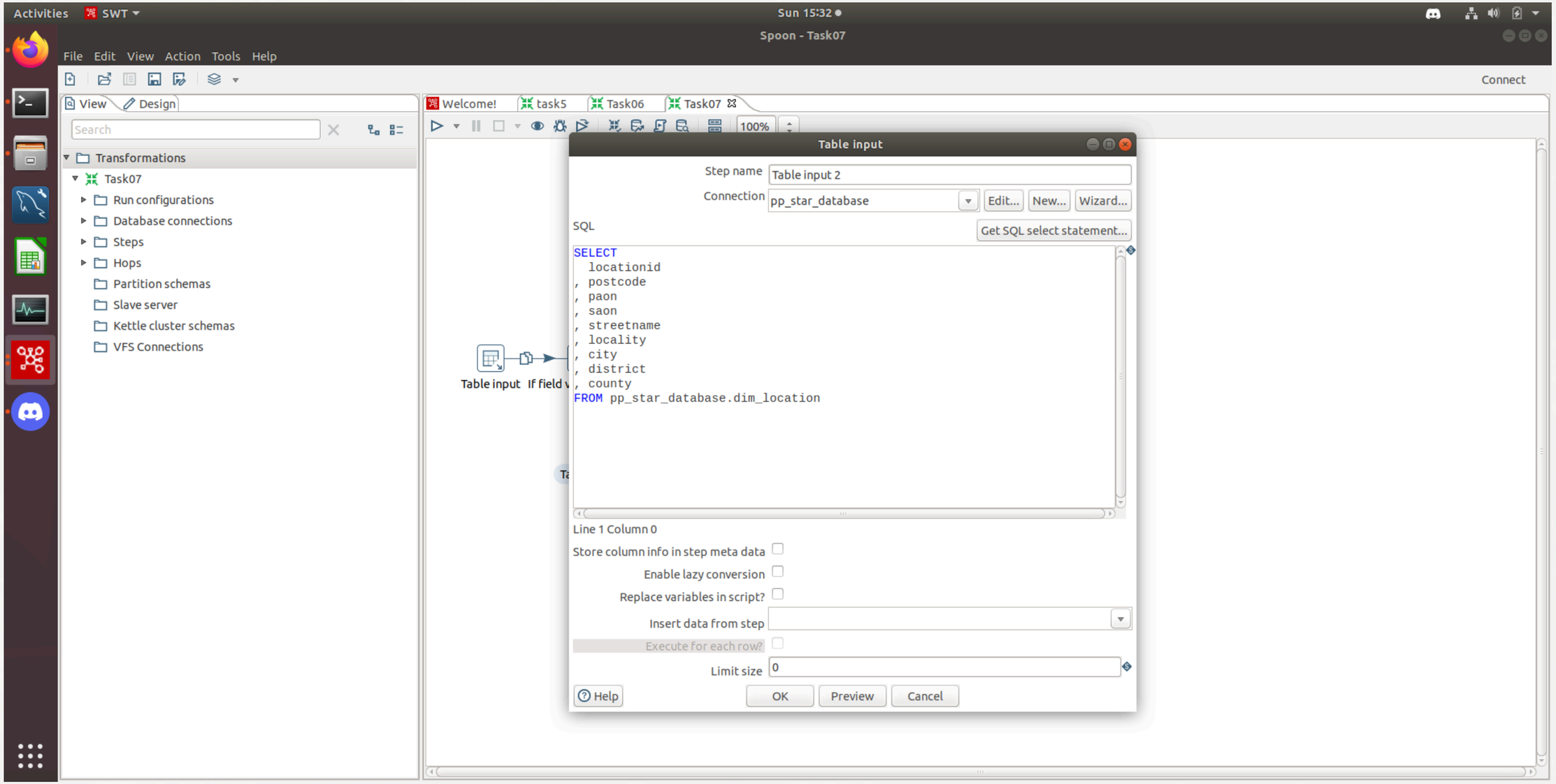


TASK 7 - Table Input (1) preview

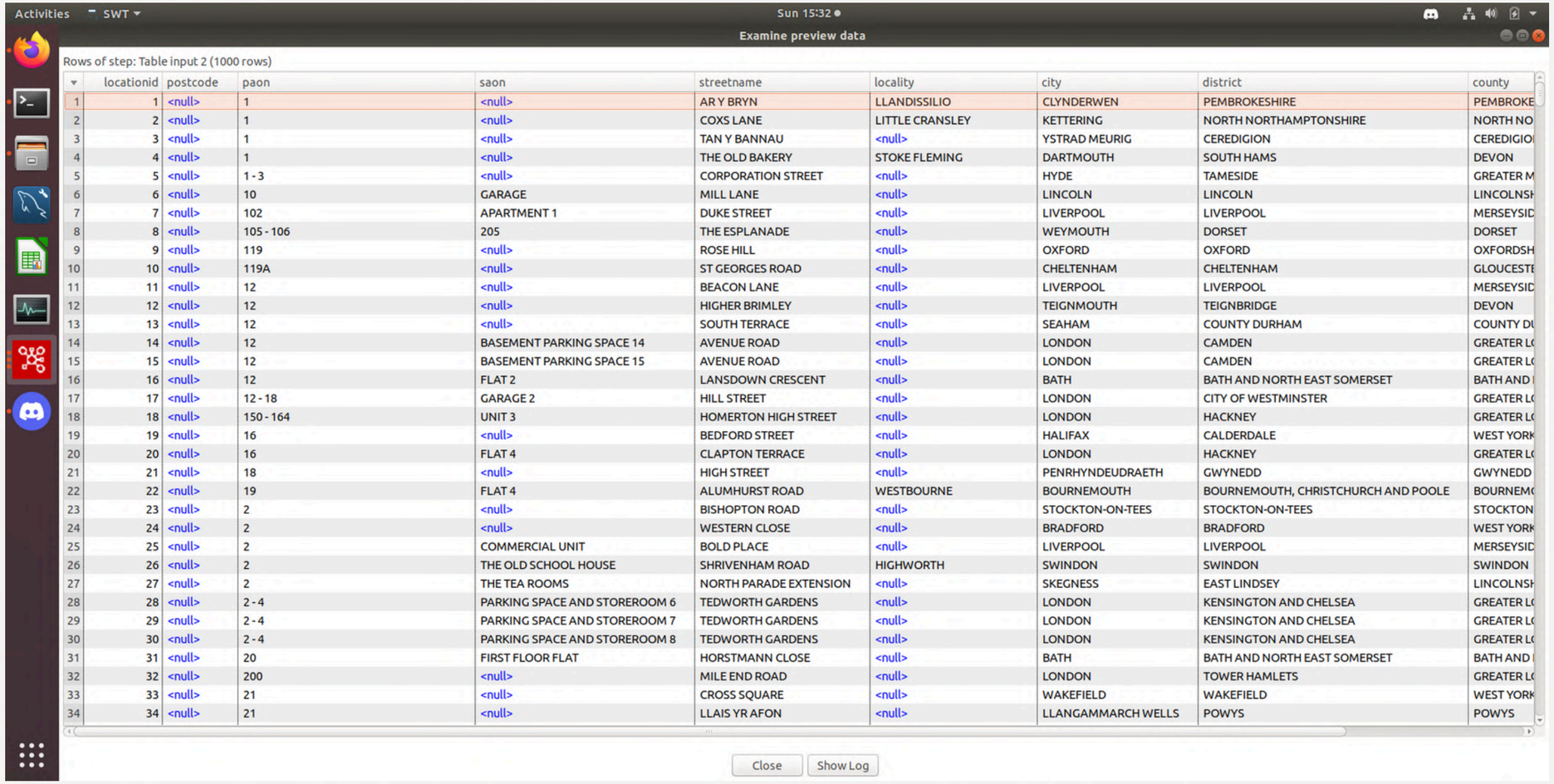


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2	{2D4D7608-8B69-BDF9-E063-4804A8C0672F}	830000	2025/01/13 00:00:00.000000000	SE13 5QQ	64		MURILLO ROAD		LONDON	LEWISHAM
3	{2D4D7608-8B6B-BDF9-E063-4804A8C0672F}	790000	2025/01/06 00:00:00.000000000	SE22 9HH	26		FELLBRIGG ROAD		LONDON	SOUTHWA
4	{2D4D7608-8B6D-BDF9-E063-4804A8C0672F}	695000	2025/01/10 00:00:00.000000000	SE13 5HW	33		WISTERIA ROAD		LONDON	LEWISHAM
5	{2D4D7608-8B70-BDF9-E063-4804A8C0672F}	1455000	2025/01/17 00:00:00.000000000	E8 4EP	250		HAGGERSTON ROAD		LONDON	HACKNEY
6	{2D4D7608-8B71-BDF9-E063-4804A8C0672F}	1955000	2025/01/10 00:00:00.000000000	SE13 7EA	20		GRANVILLE PARK		LONDON	LEWISHAM
7	{2D4D7608-8B73-BDF9-E063-4804A8C0672F}	10515000	2025/01/15 00:00:00.000000000	SE13 7EA	GRANVILLE HOUSE		GRANVILLE PARK		LONDON	LEWISHAM
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9	{2D4D7608-8B7E-BDF9-E063-4804A8C0672F}	725000	2025/01/15 00:00:00.000000000	SW16 5AZ	60		DONNYBROOK ROAD		LONDON	LAMBETH
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11	{2D4D7608-8B85-BDF9-E063-4804A8C0672F}	2700000	2025/01/06 00:00:00.000000000	N6 6EJ	4		MAKEPEACE AVENUE	HIGHGATE	LONDON	CAMDEN
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13	{2D4D7608-8B96-BDF9-E063-4804A8C0672F}	840000	2025/01/16 00:00:00.000000000	SE15 3AL	18		HICHISSON ROAD		LONDON	SOUTHWA
14	{2D4D7608-8B97-BDF9-E063-4804A8C0672F}	925000	2025/01/14 00:00:00.000000000	SE7 7EY	110		CHARLTON ROAD		LONDON	GREENWI
15	{2D4D7608-8BA2-BDF9-E063-4804A8C0672F}	770000	2025/01/24 00:00:00.000000000	SE24 0PR	145		MAYALL ROAD		LONDON	LAMBETH
16	{2D4D7608-8BA3-BDF9-E063-4804A8C0672F}	465000	2025/01/17 00:00:00.000000000	SE2 9EX	36		ABBEY GROVE	ABBEY WOOD	LONDON	GREENWI
17	{2D4D7608-8BA8-BDF9-E063-4804A8C0672F}	1760000	2025/01/01 00:00:00.000000000	SW18 5PR	66		COMBEMARTIN ROAD		LONDON	WANDSW
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21	{2D4D7608-8BB1-BDF9-E063-4804A8C0672F}	2247000	2025/01/15 00:00:00.000000000	SW6 3QD	2		BETTRIDGE ROAD		LONDON	HAMMERS
22	{2D4D7608-8BB3-BDF9-E063-4804A8C0672F}	390000	2025/01/07 00:00:00.000000000	SE2 0PB	77		WOODBROOK ROAD		LONDON	GREENWI
23	{2D4D7608-8BBC-BDF9-E063-4804A8C0672F}	550000	2025/01/20 00:00:00.000000000	SE18 3EG	9		BUSHMOOR CRESCENT		LONDON	GREENWI
24	{2D4D7608-8BC0-BDF9-E063-4804A8C0672F}	600000	2025/01/10 00:00:00.000000000	SE9 3PD	47		MONTBELLE ROAD	NEW ELTHAM	LONDON	GREENWI
25	{2D4D7608-8BC1-BDF9-E063-4804A8C0672F}	625000	2025/01/08 00:00:00.000000000	SE3 8NG	173		BROAD WALK	KIDBROOKE	LONDON	GREENWI
26	{2D4D7608-8BC4-BDF9-E063-4804A8C0672F}	970000	2025/01/23 00:00:00.000000000	SW16 6PT	1		EASTWOOD STREET		LONDON	WANDSW
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29	{2D4D7608-8BDA-BDF9-E063-4804A8C0672F}	430000	2025/01/10 00:00:00.000000000	IG9 6AS	105		LOUGHTON WAY		BUCKHURST HILL	EPPING F
30	{2D4D7608-8BDB-BDF9-E063-4804A8C0672F}	525000	2025/01/15 00:00:00.000000000	IG10 2FD	6		HIGHGROVE CLOSE		LOUGHTON	EPPING F
31	{2D4D7608-8BE4-BDF9-E063-4804A8C0672F}	340000	2025/01/21 00:00:00.000000000	CM9 6GW	1		METEOR DRIVE		MALDON	MALDON
32	{2D4D7608-8BF0-BDF9-E063-4804A8C0672F}	276500	2025/01/17 00:00:00.000000000	CO9 2GH	15		SMITH GARDENS		HALSTEAD	BRAINTRE
33	{2D4D7608-8BFD-BDF9-E063-4804A8C0672F}	565000	2025/01/23 00:00:00.000000000	CM8 1YU	21		JOSLIN AVENUE		WITHAM	BRAINTRE
34	{2D4D7608-8C07-BDF9-E063-4804A8C0672F}	295000	2025/01/13 00:00:00.000000000	SS4 1QW	1		ORMONDE AVENUE		ROCHFORD	ROCHFOR

TASK 7 - Table Input 2 configuration



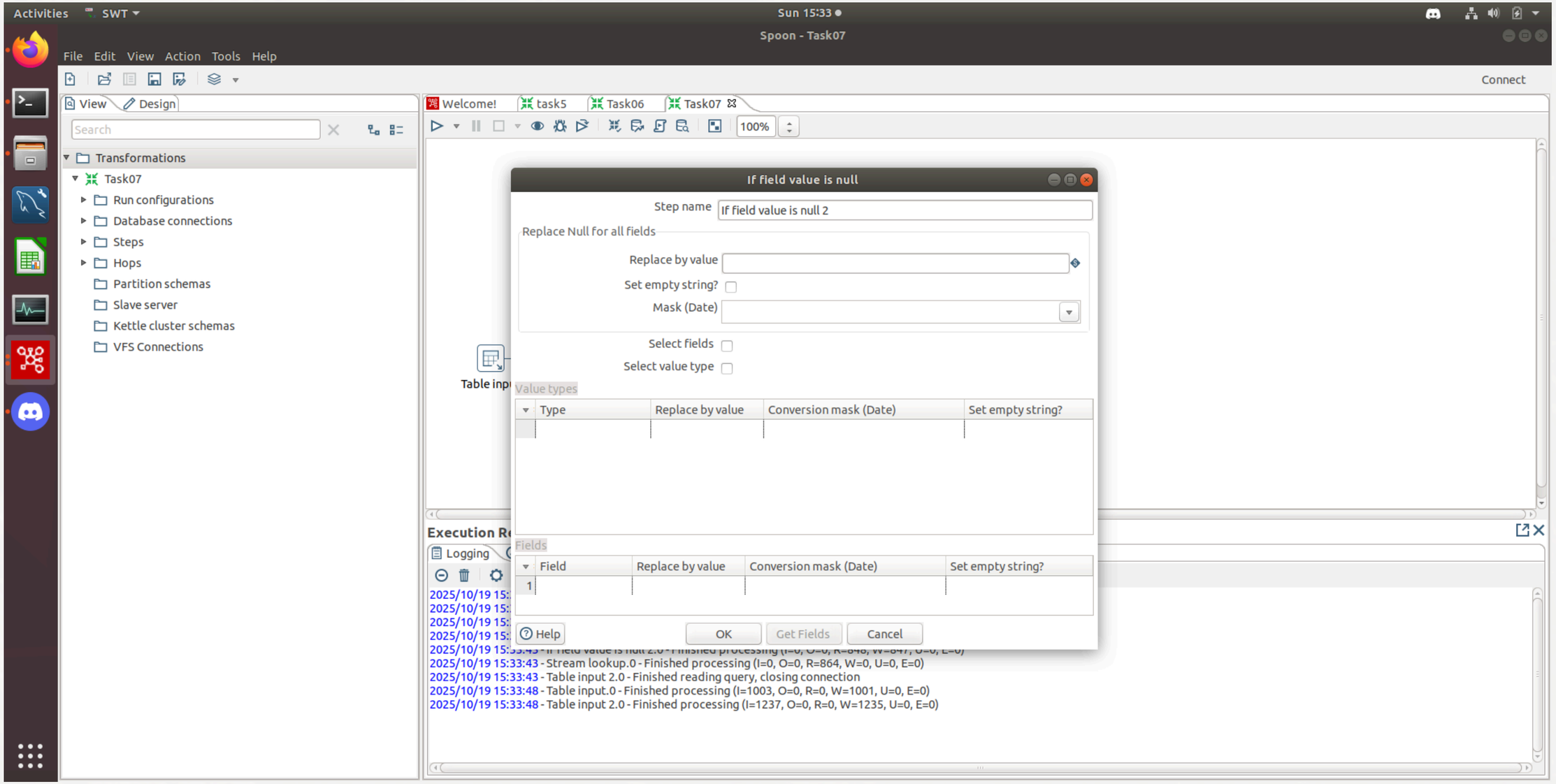
TASK 7 - Table Input 2 preview



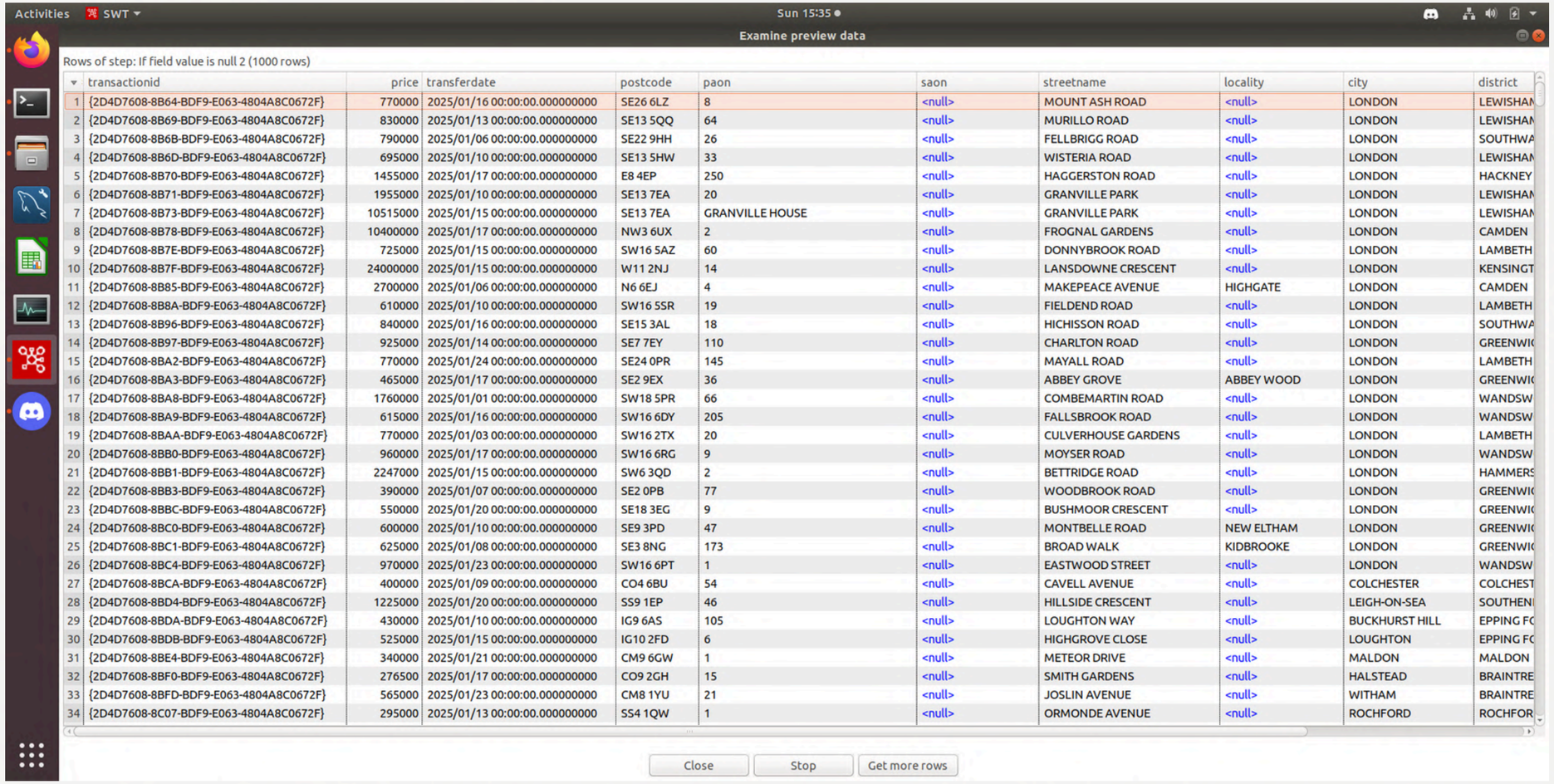
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3	3	<null>	1	<null>	TAN Y BANNAU	<null>	YSTRAD MEURIG	CEREDIGION	CEREDIGIO
4	4	<null>	1	<null>	THE OLD BAKERY	STOKE FLEMING	DARTMOUTH	SOUTH HAMS	DEVON
5	5	<null>	1 - 3	<null>	CORPORATION STREET	<null>	HYDE	TAMESIDE	GREATER M
6	6	<null>	10	GARAGE	MILL LANE	<null>	LINCOLN	LINCOLN	LINCOLNSH
7	7	<null>	102	APARTMENT 1	DUKE STREET	<null>	LIVERPOOL	LIVERPOOL	MERSEYSID
8	8	<null>	105 - 106	205	THE ESPLANADE	<null>	WEYMOUTH	DORSET	DORSET
9	9	<null>	119	<null>	ROSE HILL	<null>	OXFORD	OXFORD	OXFORDSH
10	10	<null>	119A	<null>	ST GEORGES ROAD	<null>	CHELTENHAM	CHELTENHAM	GLOUCESTE
11	11	<null>	12	<null>	BEACON LANE	<null>	LIVERPOOL	LIVERPOOL	MERSEYSID
12	12	<null>	12	<null>	HIGHER BRIMLEY	<null>	TEIGNMOUTH	TEIGNBRIDGE	DEVON
13	13	<null>	12	<null>	SOUTH TERRACE	<null>	SEAHAM	COUNTY DURHAM	COUNTY DU
14	14	<null>	12	BASEMENT PARKING SPACE 14	AVENUE ROAD	<null>	LONDON	CAMDEN	GREATER LO
15	15	<null>	12	BASEMENT PARKING SPACE 15	AVENUE ROAD	<null>	LONDON	CAMDEN	GREATER LO
16	16	<null>	12	FLAT 2	LANSDOWN CRESCENT	<null>	BATH	BATH AND NORTH EAST SOMERSET	BATH AND
17	17	<null>	12 - 18	GARAGE 2	HILL STREET	<null>	LONDON	CITY OF WESTMINSTER	GREATER LO
18	18	<null>	150 - 164	UNIT 3	HOMERTON HIGH STREET	<null>	LONDON	HACKNEY	GREATER LO
19	19	<null>	16	<null>	BEDFORD STREET	<null>	HALIFAX	CALDERDALE	WEST YORK
20	20	<null>	16	FLAT 4	CLAPTON TERRACE	<null>	LONDON	HACKNEY	GREATER LO
21	21	<null>	18	<null>	HIGH STREET	<null>	PENRHYNDEUDRAETH	GWYNEDD	GWYNEDD
22	22	<null>	19	FLAT 4	ALUMHURST ROAD	WESTBOURNE	BOURNEMOUTH	BOURNEMOUTH, CHRISTCHURCH AND POOLE	BOURNEMO
23	23	<null>	2	<null>	BISHOPTON ROAD	<null>	STOCKTON-ON-TEES	STOCKTON-ON-TEES	STOCKTON
24	24	<null>	2	<null>	WESTERN CLOSE	<null>	BRADFORD	BRADFORD	WEST YORK
25	25	<null>	2	COMMERCIAL UNIT	BOLD PLACE	<null>	LIVERPOOL	LIVERPOOL	MERSEYSID
26	26	<null>	2	THE OLD SCHOOL HOUSE	SHRIVENHAM ROAD	HIGHWORTH	SWINDON	SWINDON	SWINDON
27	27	<null>	2	THE TEA ROOMS	NORTH PARADE EXTENSION	<null>	SKEGNESS	EAST LINDSEY	LINCOLNSH
28	28	<null>	2 - 4	PARKING SPACE AND STOREROOM 6	TEDWORTH GARDENS	<null>	LONDON	KENSINGTON AND CHELSEA	GREATER LO
29	29	<null>	2 - 4	PARKING SPACE AND STOREROOM 7	TEDWORTH GARDENS	<null>	LONDON	KENSINGTON AND CHELSEA	GREATER LO
30	30	<null>	2 - 4	PARKING SPACE AND STOREROOM 8	TEDWORTH GARDENS	<null>	LONDON	KENSINGTON AND CHELSEA	GREATER LO
31	31	<null>	20	FIRST FLOOR FLAT	HORSTMANN CLOSE	<null>	BATH	BATH AND NORTH EAST SOMERSET	BATH AND
32	32	<null>	200	<null>	MILE END ROAD	<null>	LONDON	TOWER HAMLETS	GREATER LO
33	33	<null>	21	<null>	CROSS SQUARE	<null>	WAKEFIELD	WAKEFIELD	WEST YORK
34	34	<null>	21	<null>	LLAIS YR AFON	<null>	LLANGAMMARCH WELLS	POWYS	POWYS

Note: the empty strings/values from previous transformations seem to have been automatically transformed into NULLs when loaded into the tables. We will deal with this by using “If field value is null” steps to transform them back into empty values (for comparisons)

TASK 7 - If value is null 2 - configuration



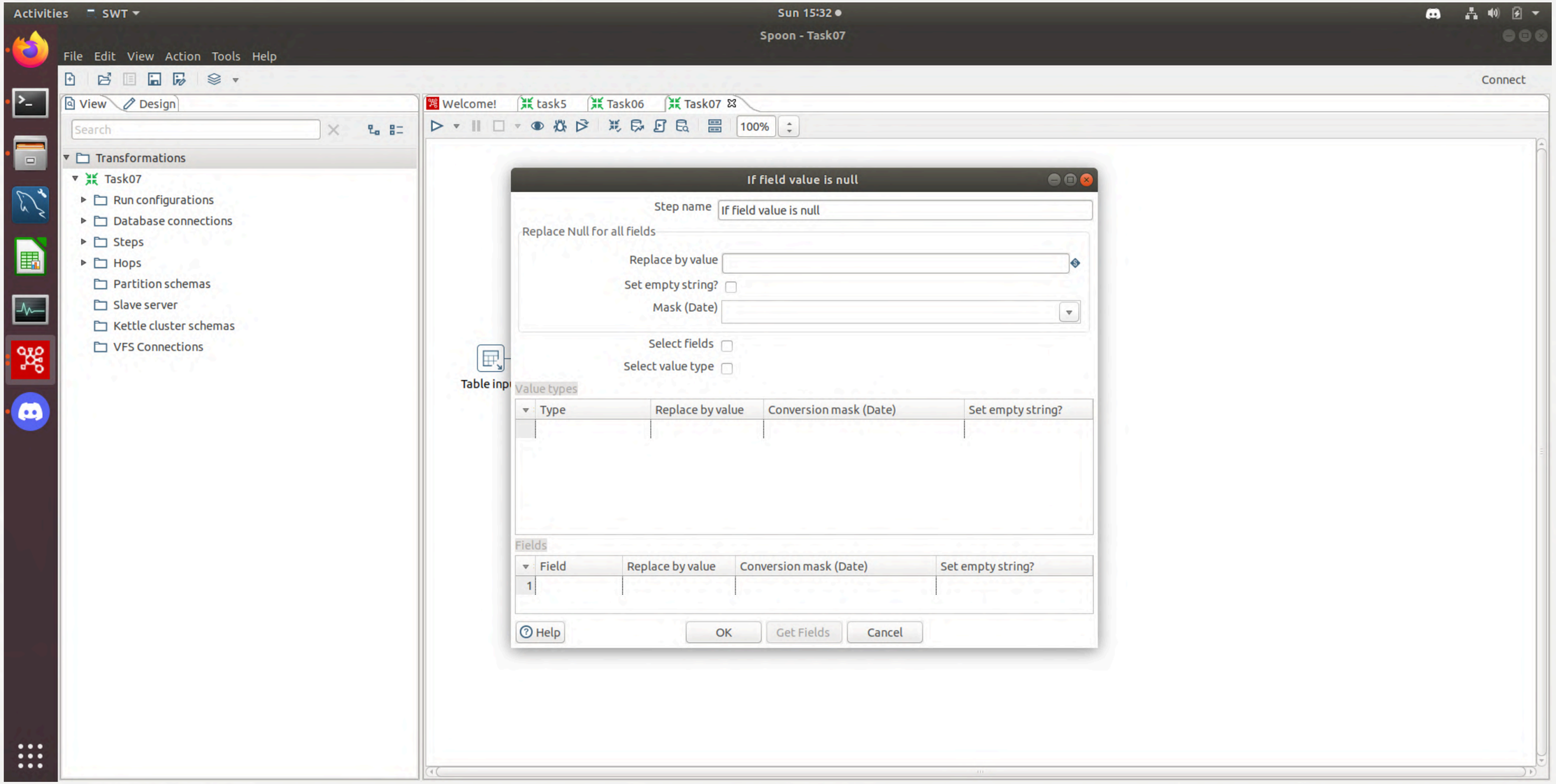
TASK 7 - If value is null 2 - preview



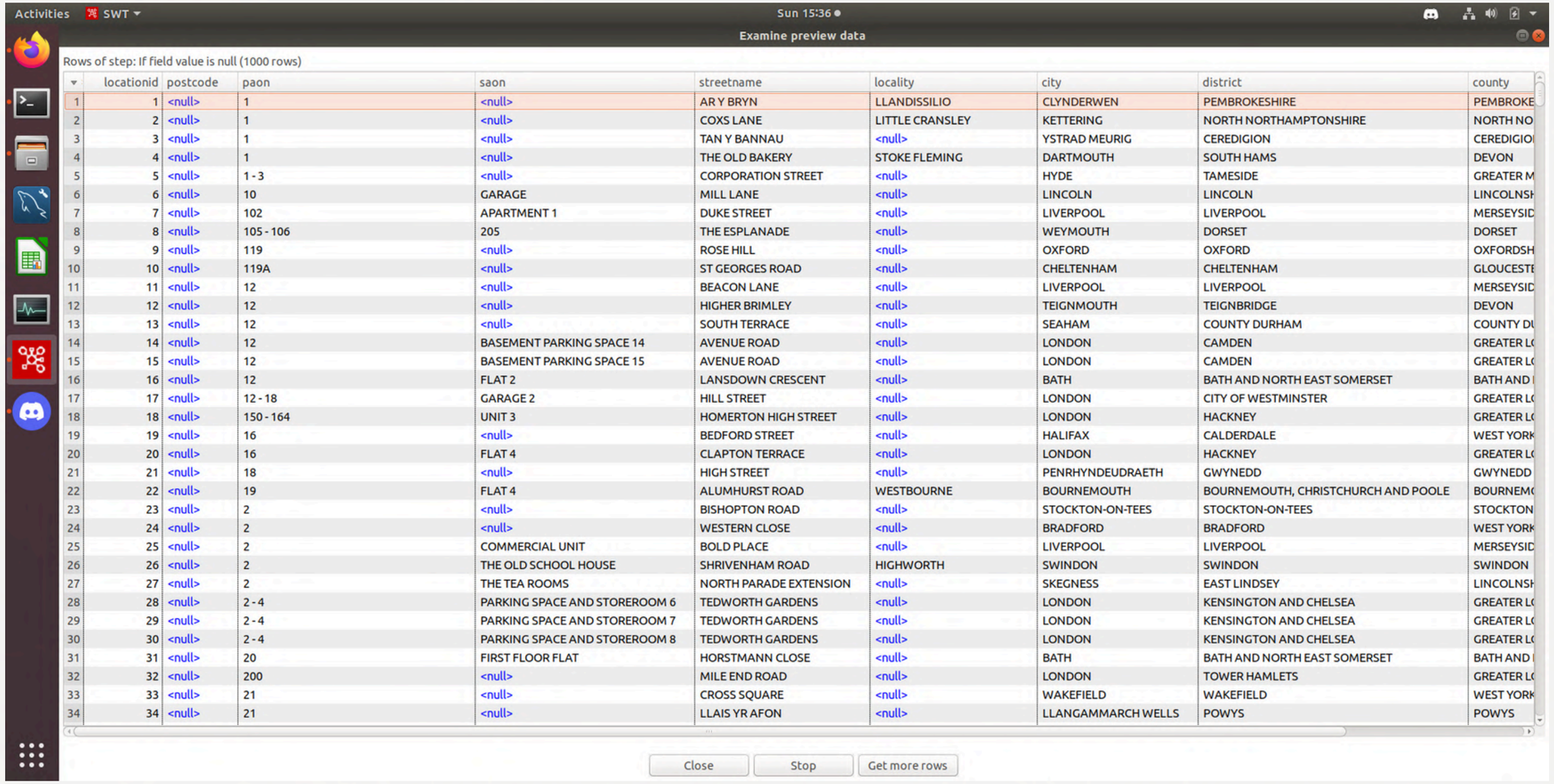
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2	{2D4D7608-8B69-BDF9-E063-4804A8C0672F}	830000	2025/01/13 00:00:00.000000000	SE13 5QQ	64	<null>	MURILLO ROAD	<null>	LONDON	LEWISHAM
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4	{2D4D7608-8B6D-BDF9-E063-4804A8C0672F}	695000	2025/01/10 00:00:00.000000000	SE13 5HW	33	<null>	WISTERIA ROAD	<null>	LONDON	LEWISHAM
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Note: from the preview, it appears as if the step transformed all empty strings/values into null, and not the other way around. The final result seems to be correct, however.

TASK 7 - If value is null (1) - configuration



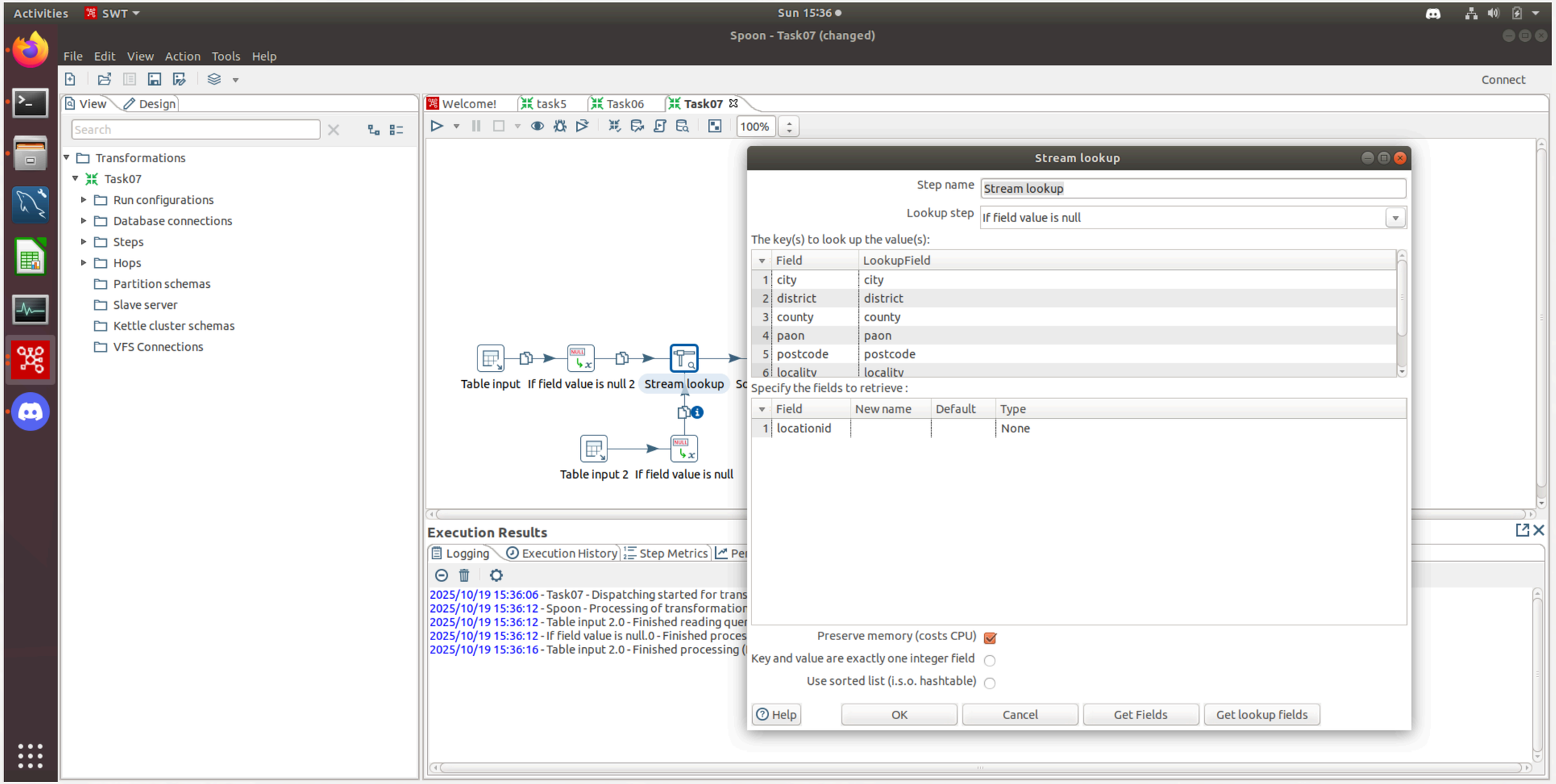
TASK 7 - If value is null (1) preview



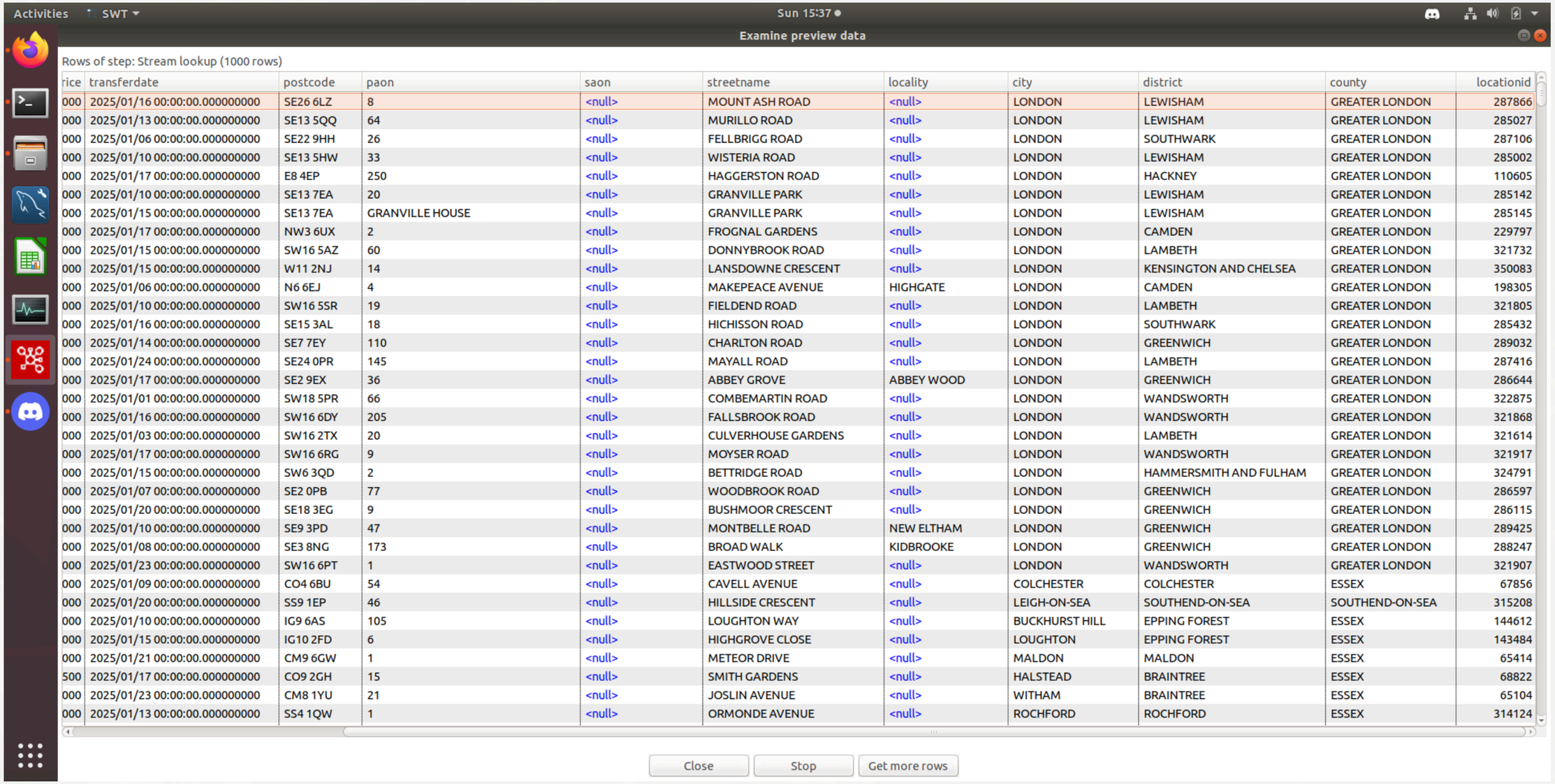
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5	5	<null>	1 - 3	<null>	CORPORATION STREET	<null>	HYDE	TAMESIDE	GREATER M
6	6	<null>	10	GARAGE	MILL LANE	<null>	LINCOLN	LINCOLN	LINCOLNSH
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Note: from the preview, it appears as if the step transformed all empty strings/values into null and not the other way around. The final result seems to be correct, however.

TASK 7 - Stream Lookup configuration

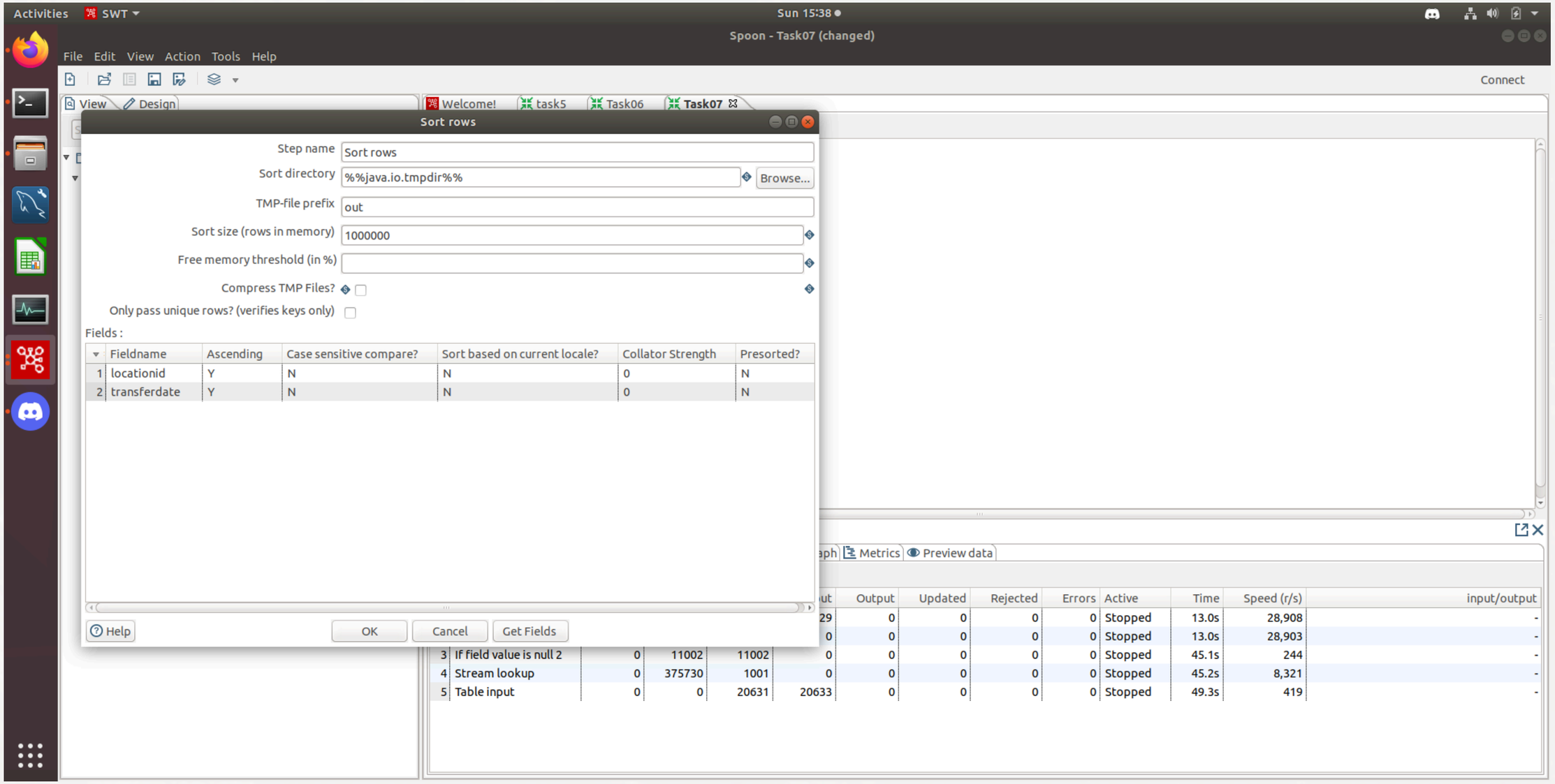


TASK 7 - Stream Lookup preview

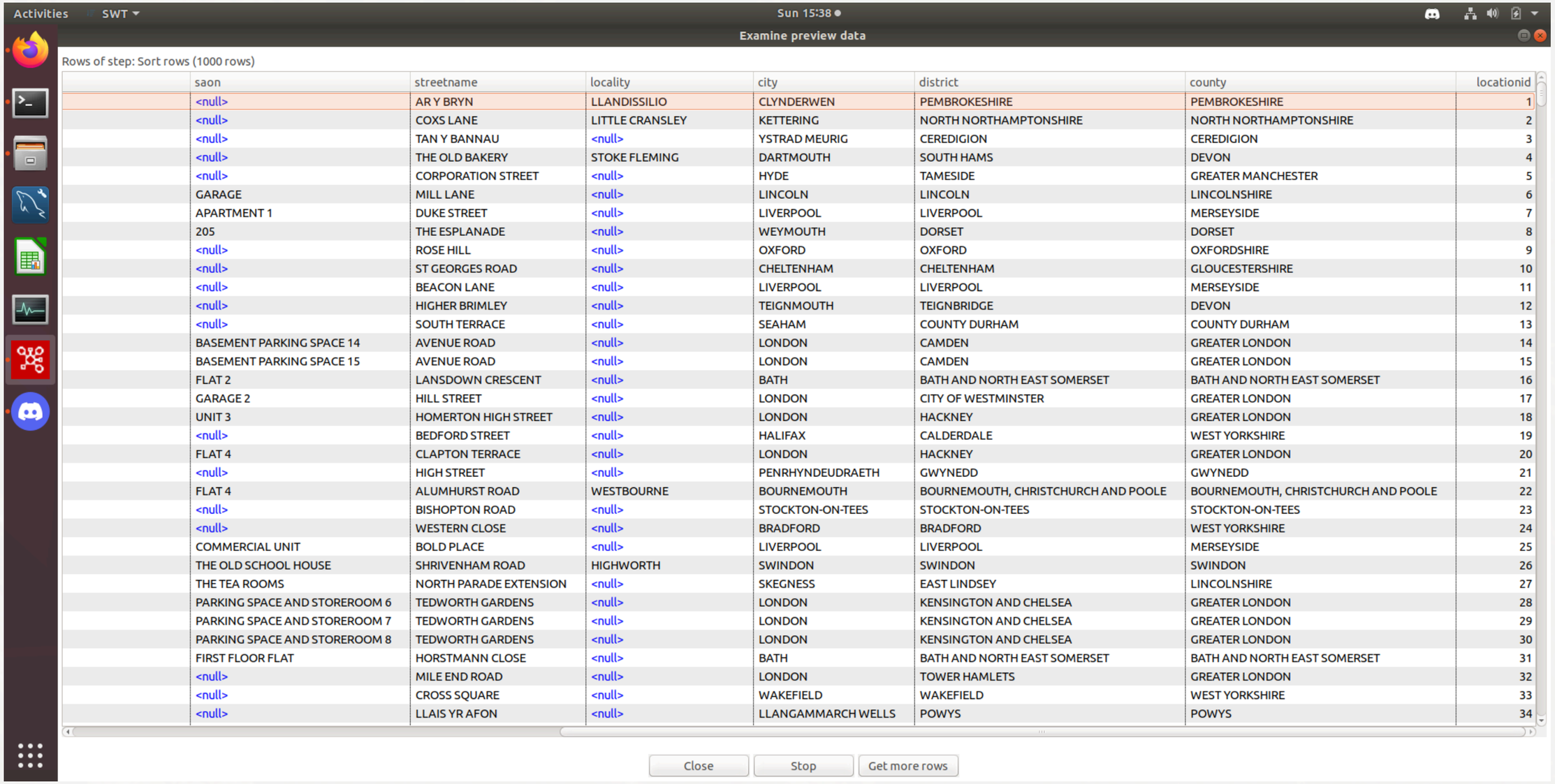


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TASK 7 - Sort Rows configuration

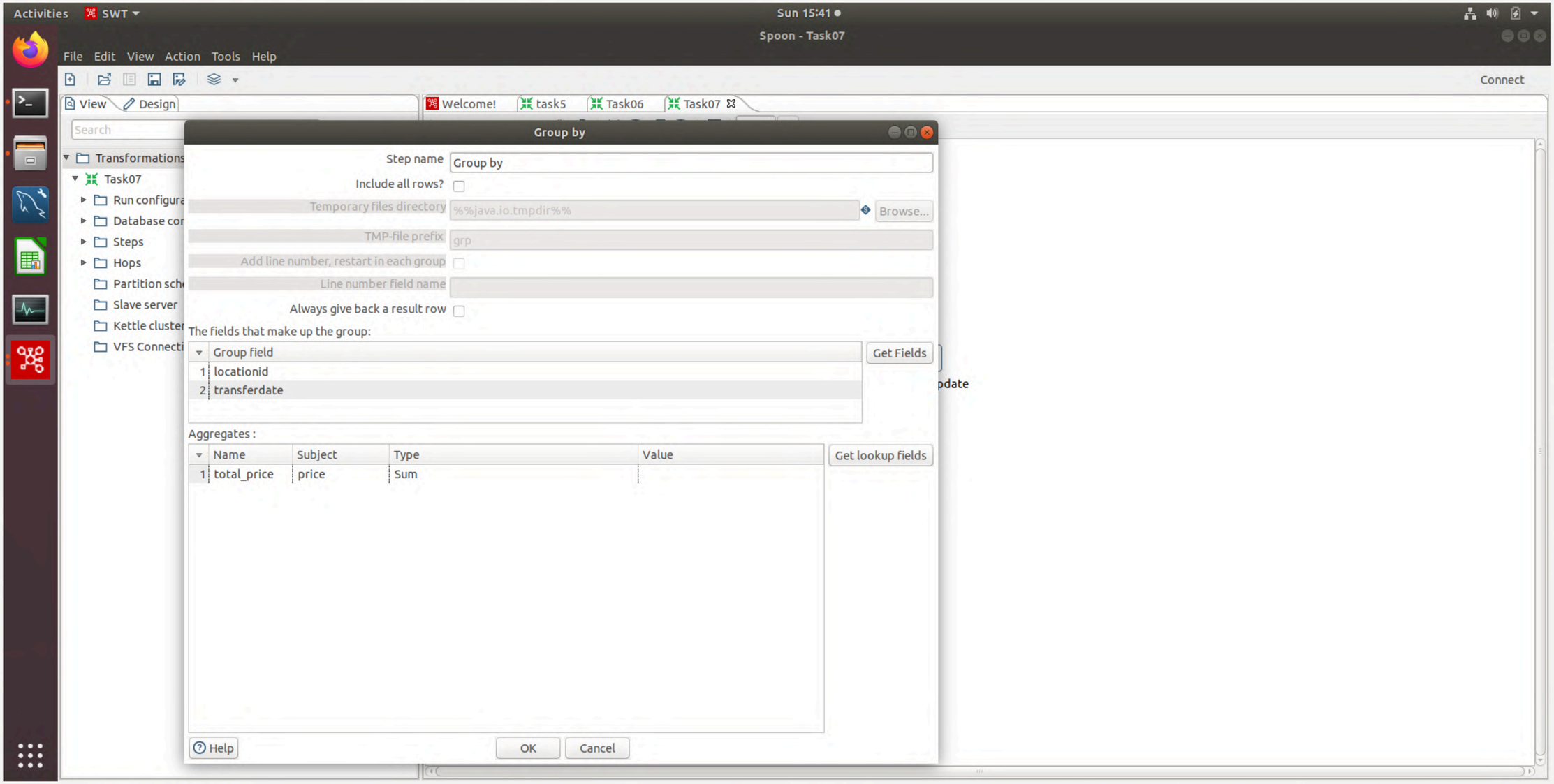


TASK 7 - Sort Rows preview

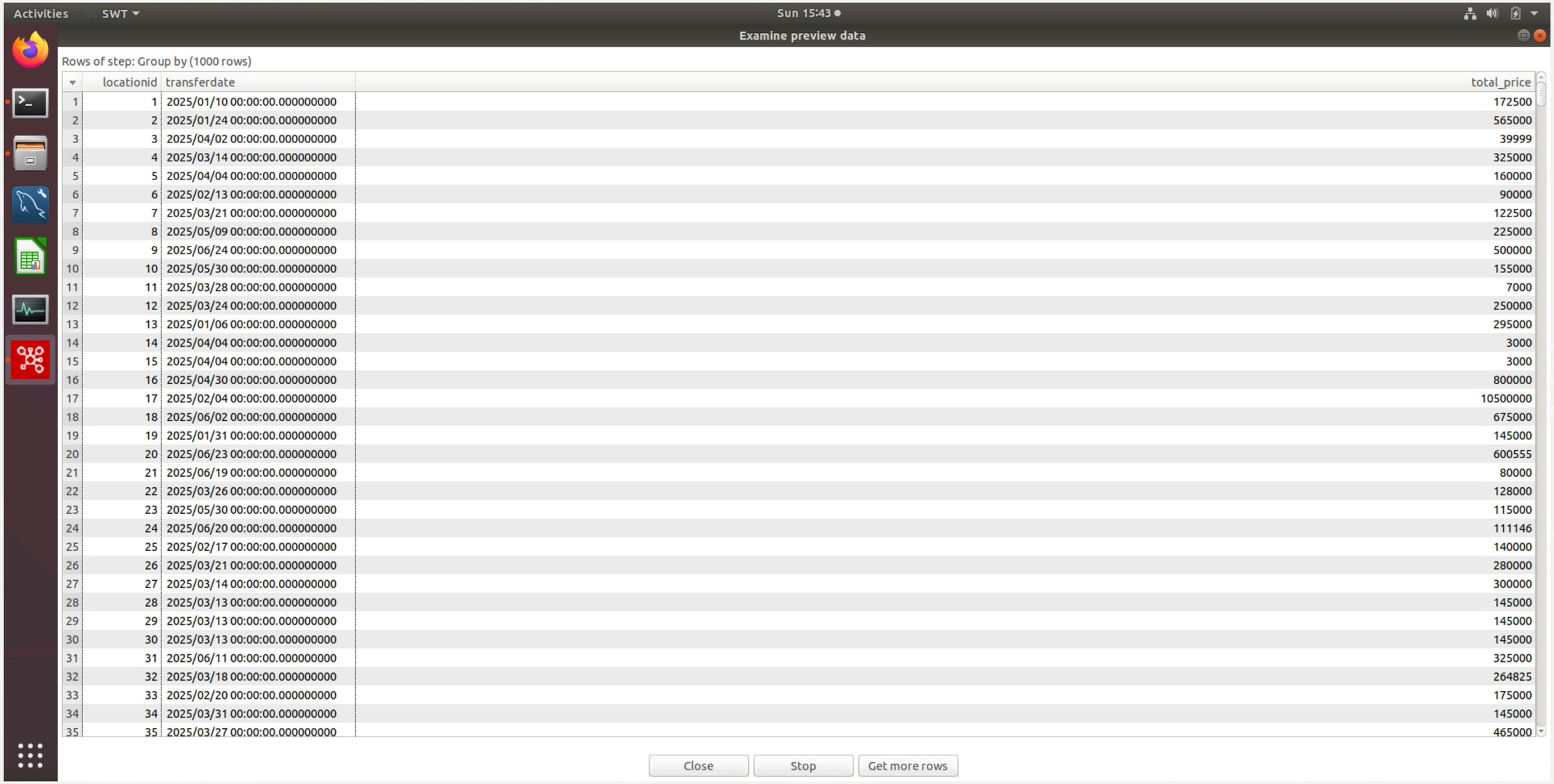


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	COMMERCIAL UNIT	BOLD PLACE	<null>	LIVERPOOL	LIVERPOOL	MERSEYSIDE	25
	THE OLD SCHOOL HOUSE	SHRIVENHAM ROAD	HIGHWORTH	SWINDON	SWINDON	SWINDON	26
	THE TEA ROOMS	NORTH PARADE EXTENSION	<null>	SKEGNESS	EAST LINDSEY	LINCOLNSHIRE	27
	PARKING SPACE AND STOREROOM 6	TEDWORTH GARDENS	<null>	LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	28
	PARKING SPACE AND STOREROOM 7	TEDWORTH GARDENS	<null>	LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	29
	PARKING SPACE AND STOREROOM 8	TEDWORTH GARDENS	<null>	LONDON	KENSINGTON AND CHELSEA	GREATER LONDON	30
	FIRST FLOOR FLAT	HORSTMANN CLOSE	<null>	BATH	BATH AND NORTH EAST SOMERSET	BATH AND NORTH EAST SOMERSET	31
	<null>	MILE END ROAD	<null>	LONDON	TOWER HAMLETS	GREATER LONDON	32
	<null>	CROSS SQUARE	<null>	WAKEFIELD	WAKEFIELD	WEST YORKSHIRE	33
	<null>	LLAIS YR AFON	<null>	LLANGAMMARCH WELLS	POWYS	POWYS	34

TASK 7 - Group By configuration

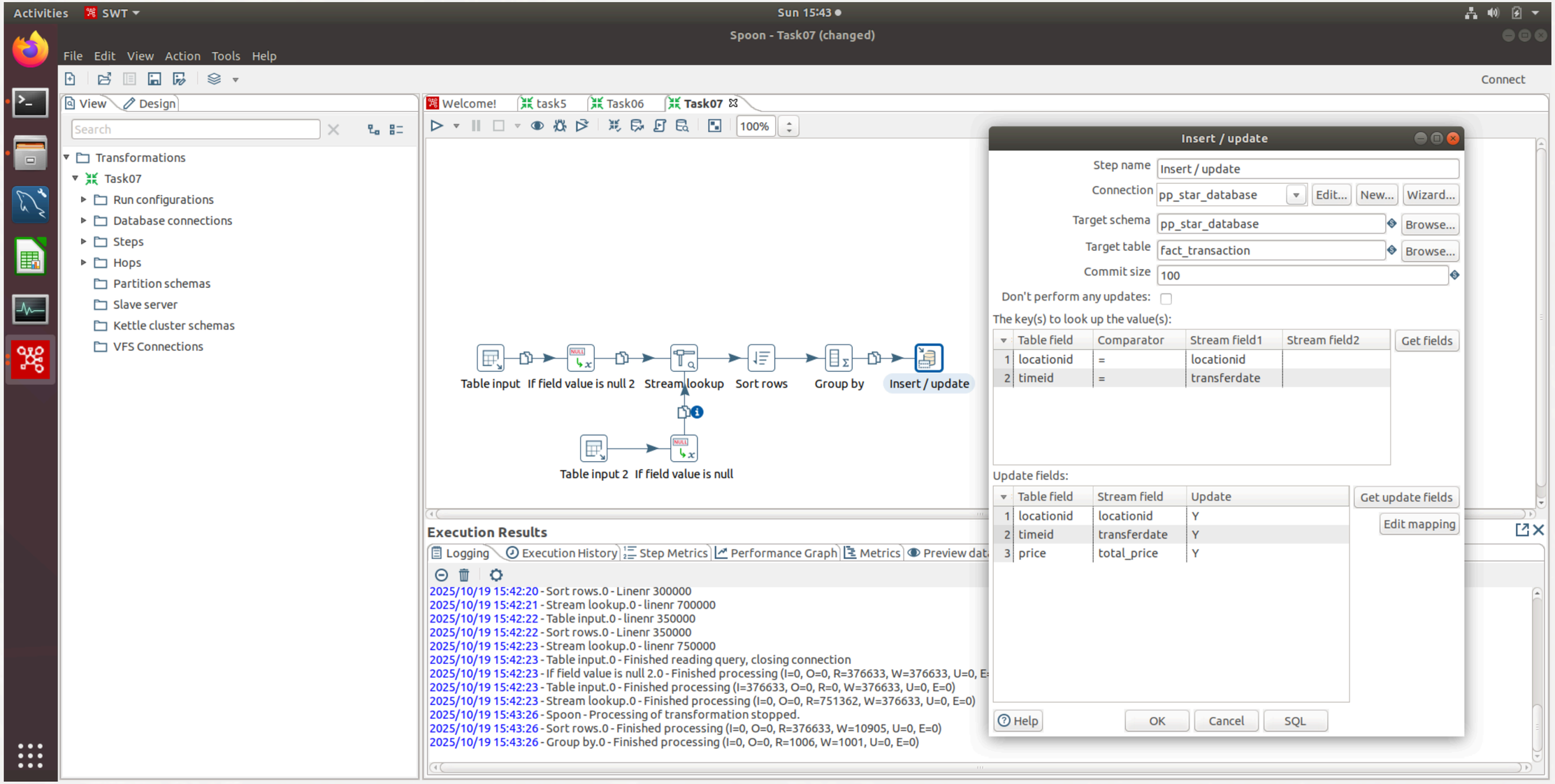


TASK 7 - Group By preview

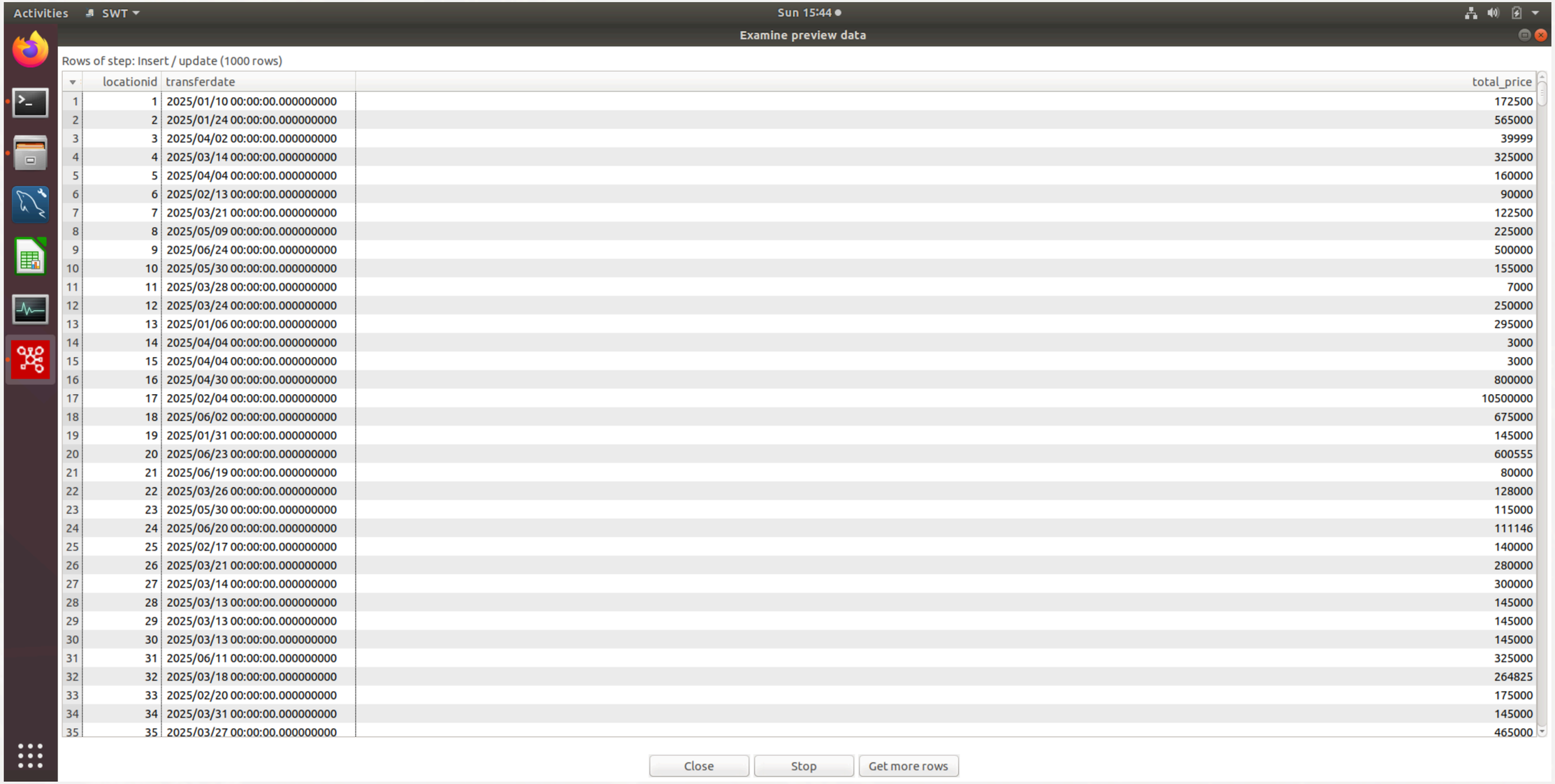


	locationid	transferdate	total_price
1	1	2025/01/10 00:00:00.000000000	172500
2	2	2025/01/24 00:00:00.000000000	565000
3	3	2025/04/02 00:00:00.000000000	39999
4	4	2025/03/14 00:00:00.000000000	325000
5	5	2025/04/04 00:00:00.000000000	160000
6	6	2025/02/13 00:00:00.000000000	90000
7	7	2025/03/21 00:00:00.000000000	122500
8	8	2025/05/09 00:00:00.000000000	225000
9	9	2025/06/24 00:00:00.000000000	500000
10	10	2025/05/30 00:00:00.000000000	155000
11	11	2025/03/28 00:00:00.000000000	7000
12	12	2025/03/24 00:00:00.000000000	250000
13	13	2025/01/06 00:00:00.000000000	295000
14	14	2025/04/04 00:00:00.000000000	3000
15	15	2025/04/04 00:00:00.000000000	3000
16	16	2025/04/30 00:00:00.000000000	800000
17	17	2025/02/04 00:00:00.000000000	10500000
18	18	2025/06/02 00:00:00.000000000	675000
19	19	2025/01/31 00:00:00.000000000	145000
20	20	2025/06/23 00:00:00.000000000	600555
21	21	2025/06/19 00:00:00.000000000	80000
22	22	2025/03/26 00:00:00.000000000	128000
23	23	2025/05/30 00:00:00.000000000	115000
24	24	2025/06/20 00:00:00.000000000	111146
25	25	2025/02/17 00:00:00.000000000	140000
26	26	2025/03/21 00:00:00.000000000	280000
27	27	2025/03/14 00:00:00.000000000	300000
28	28	2025/03/13 00:00:00.000000000	145000
29	29	2025/03/13 00:00:00.000000000	145000
30	30	2025/03/13 00:00:00.000000000	145000
31	31	2025/06/11 00:00:00.000000000	325000
32	32	2025/03/18 00:00:00.000000000	264825
33	33	2025/02/20 00:00:00.000000000	175000
34	34	2025/03/31 00:00:00.000000000	145000
35	35	2025/03/27 00:00:00.000000000	465000

TASK 7 - Insert/Update configuration

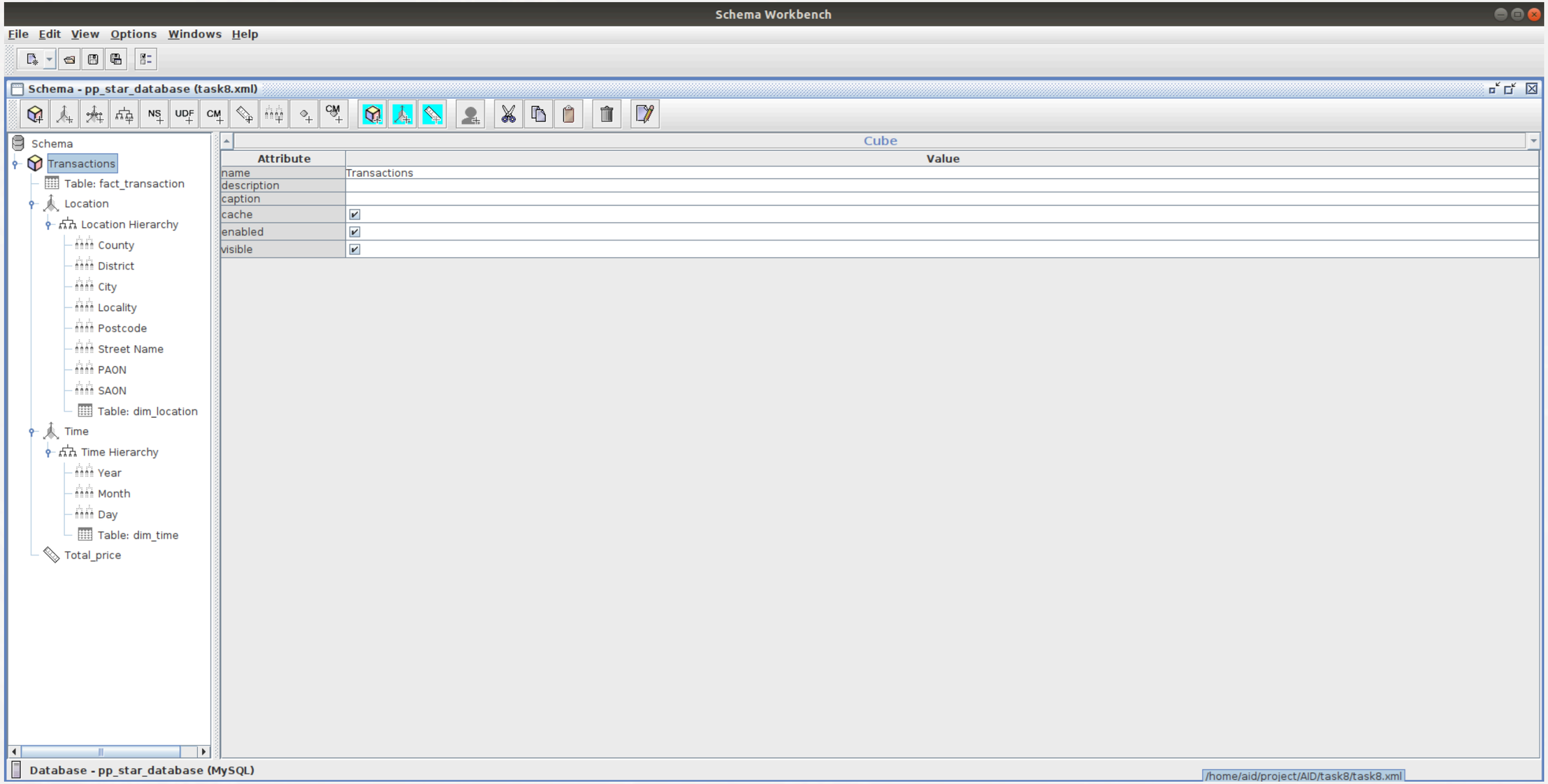


TASK 7 - Insert/Update preview

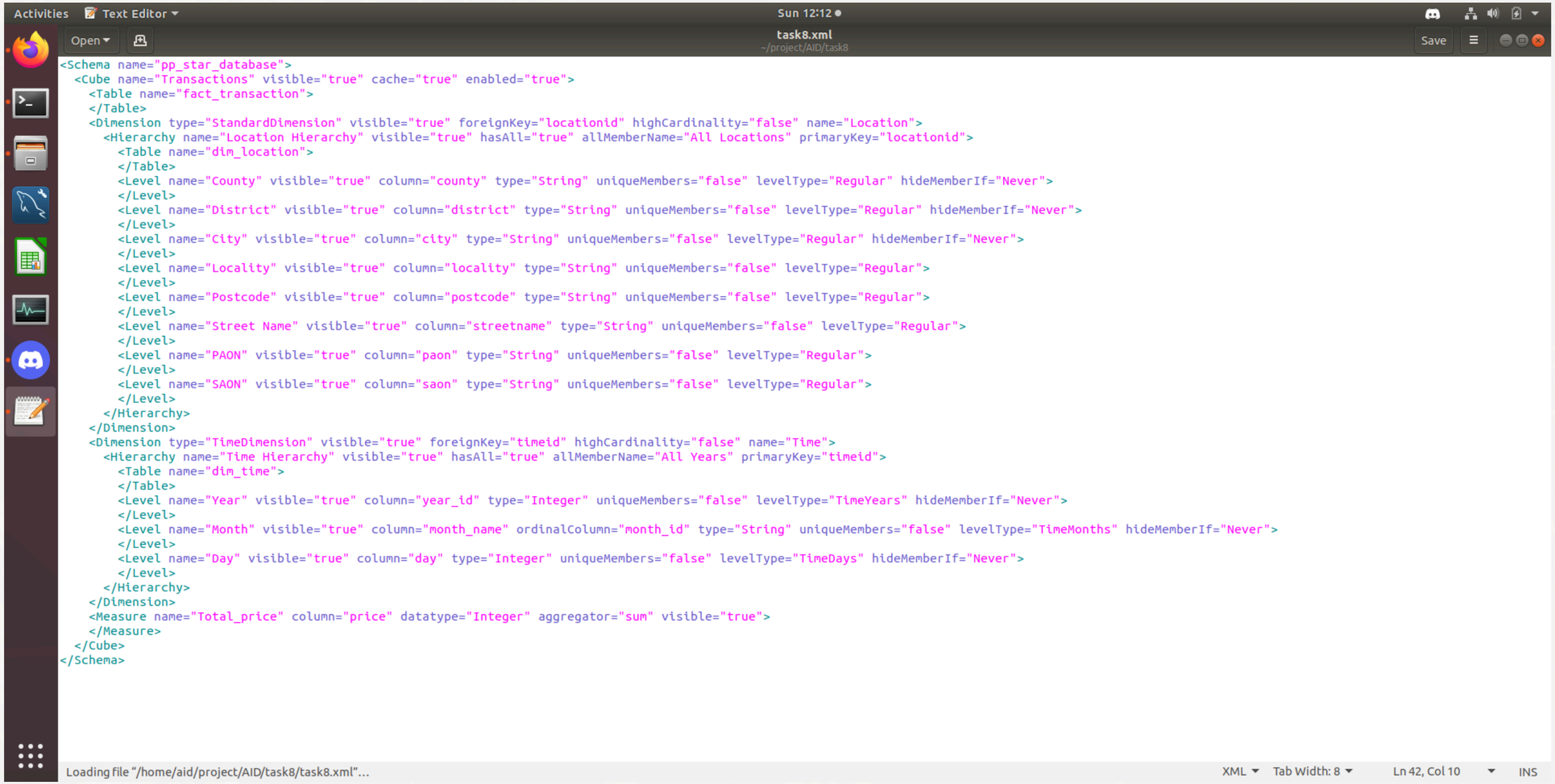


	locationid	transferdate	total_price
1	1	2025/01/10 00:00:00.000000000	172500
2	2	2025/01/24 00:00:00.000000000	565000
3	3	2025/04/02 00:00:00.000000000	39999
4	4	2025/03/14 00:00:00.000000000	325000
5	5	2025/04/04 00:00:00.000000000	160000
6	6	2025/02/13 00:00:00.000000000	90000
7	7	2025/03/21 00:00:00.000000000	122500
8	8	2025/05/09 00:00:00.000000000	225000
9	9	2025/06/24 00:00:00.000000000	500000
10	10	2025/05/30 00:00:00.000000000	155000
11	11	2025/03/28 00:00:00.000000000	7000
12	12	2025/03/24 00:00:00.000000000	250000
13	13	2025/01/06 00:00:00.000000000	295000
14	14	2025/04/04 00:00:00.000000000	3000
15	15	2025/04/04 00:00:00.000000000	3000
16	16	2025/04/30 00:00:00.000000000	800000
17	17	2025/02/04 00:00:00.000000000	10500000
18	18	2025/06/02 00:00:00.000000000	675000
19	19	2025/01/31 00:00:00.000000000	145000
20	20	2025/06/23 00:00:00.000000000	600555
21	21	2025/06/19 00:00:00.000000000	80000
22	22	2025/03/26 00:00:00.000000000	128000
23	23	2025/05/30 00:00:00.000000000	115000
24	24	2025/06/20 00:00:00.000000000	111146
25	25	2025/02/17 00:00:00.000000000	140000
26	26	2025/03/21 00:00:00.000000000	280000
27	27	2025/03/14 00:00:00.000000000	300000
28	28	2025/03/13 00:00:00.000000000	145000
29	29	2025/03/13 00:00:00.000000000	145000
30	30	2025/03/13 00:00:00.000000000	145000
31	31	2025/06/11 00:00:00.000000000	325000
32	32	2025/03/18 00:00:00.000000000	264825
33	33	2025/02/20 00:00:00.000000000	175000
34	34	2025/03/31 00:00:00.000000000	145000
35	35	2025/03/27 00:00:00.000000000	465000

TASK 8 - Pentaho Schema Workbench Tree

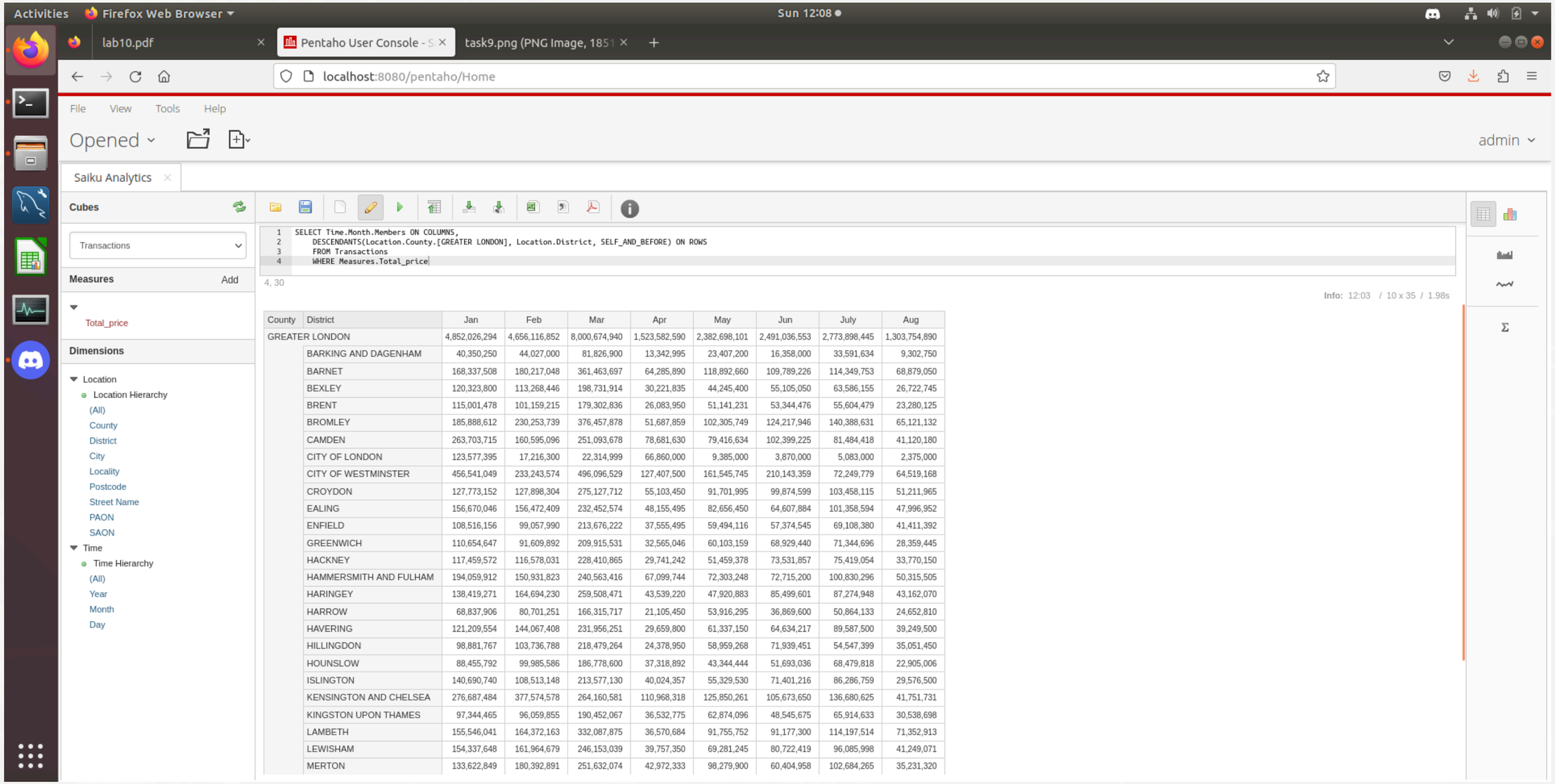


TASK 8 - XML Contents



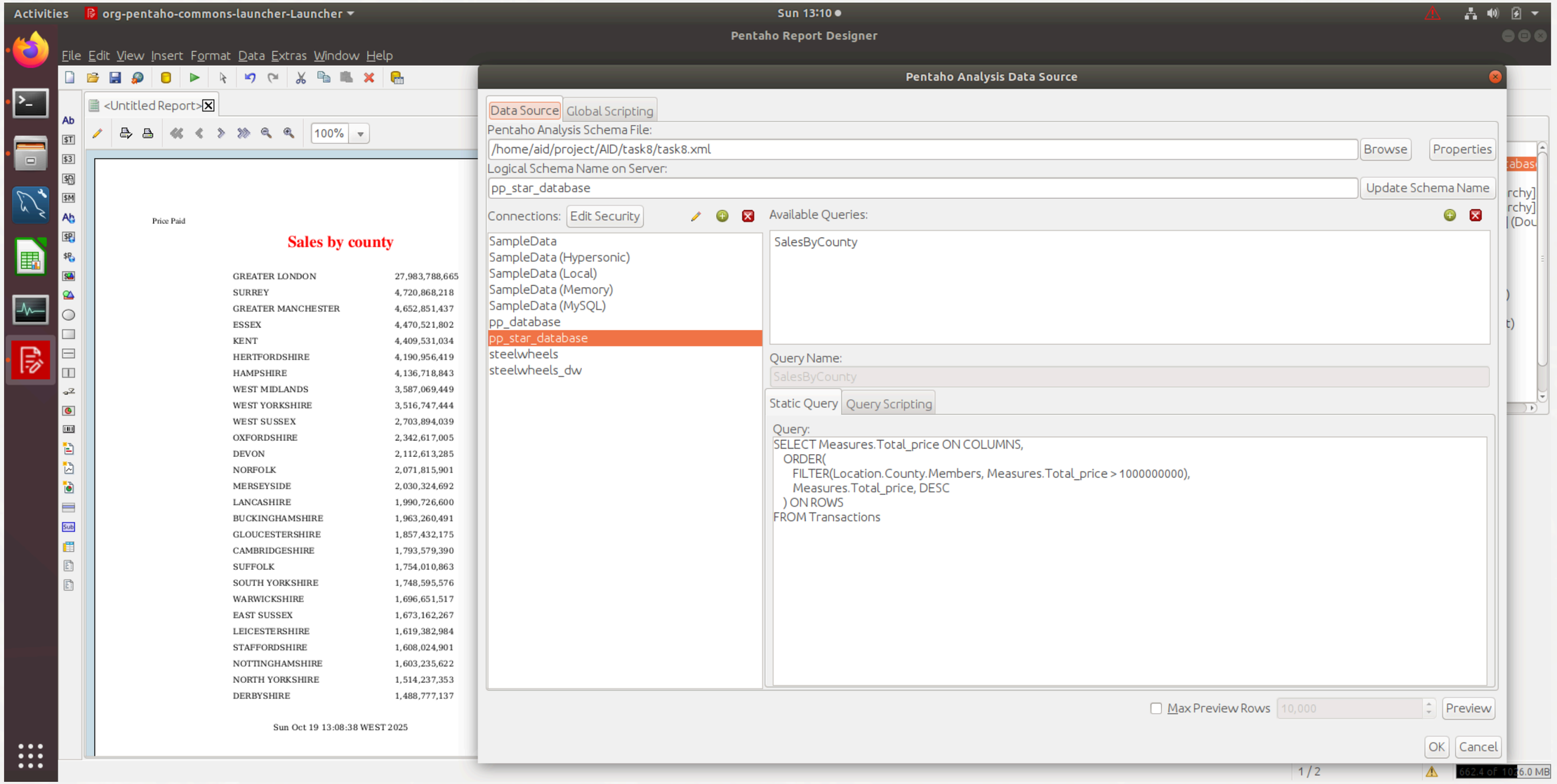

```xml
<Schema name="pp_star_database">
  <Cube name="Transactions" visible="true" cache="true" enabled="true">
    <Table name="fact_transaction">
    </Table>
    <Dimension type="StandardDimension" visible="true" foreignKey="locationid" highCardinality="false" name="Location">
      <Hierarchy name="Location Hierarchy" visible="true" hasAll="true" allMemberName="All Locations" primaryKey="locationid">
        <Table name="dim_location">
        </Table>
        <Level name="County" visible="true" column="county" type="String" uniqueMembers="false" levelType="Regular" hideMemberIf="Never">
        </Level>
        <Level name="District" visible="true" column="district" type="String" uniqueMembers="false" levelType="Regular" hideMemberIf="Never">
        </Level>
        <Level name="City" visible="true" column="city" type="String" uniqueMembers="false" levelType="Regular" hideMemberIf="Never">
        </Level>
        <Level name="Locality" visible="true" column="locality" type="String" uniqueMembers="false" levelType="Regular">
        </Level>
        <Level name="Postcode" visible="true" column="postcode" type="String" uniqueMembers="false" levelType="Regular">
        </Level>
        <Level name="Street Name" visible="true" column="streetname" type="String" uniqueMembers="false" levelType="Regular">
        </Level>
        <Level name="PAON" visible="true" column="paon" type="String" uniqueMembers="false" levelType="Regular">
        </Level>
        <Level name="SAON" visible="true" column="saon" type="String" uniqueMembers="false" levelType="Regular">
        </Level>
      </Hierarchy>
    </Dimension>
    <Dimension type="TimeDimension" visible="true" foreignKey="timeid" highCardinality="false" name="Time">
      <Hierarchy name="Time Hierarchy" visible="true" hasAll="true" allMemberName="All Years" primaryKey="timeid">
        <Table name="dim_time">
        </Table>
        <Level name="Year" visible="true" column="year_id" type="Integer" uniqueMembers="false" levelType="TimeYears" hideMemberIf="Never">
        </Level>
        <Level name="Month" visible="true" column="month_name" ordinalColumn="month_id" type="String" uniqueMembers="false" levelType="TimeMonths" hideMemberIf="Never">
        </Level>
        <Level name="Day" visible="true" column="day" type="Integer" uniqueMembers="false" levelType="TimeDays" hideMemberIf="Never">
        </Level>
      </Hierarchy>
    </Dimension>
    <Measure name="Total_price" column="price" datatype="Integer" aggregator="sum" visible="true">
    </Measure>
  </Cube>
</Schema>
```

TASK 9 - Saiku Analytics analysis and results

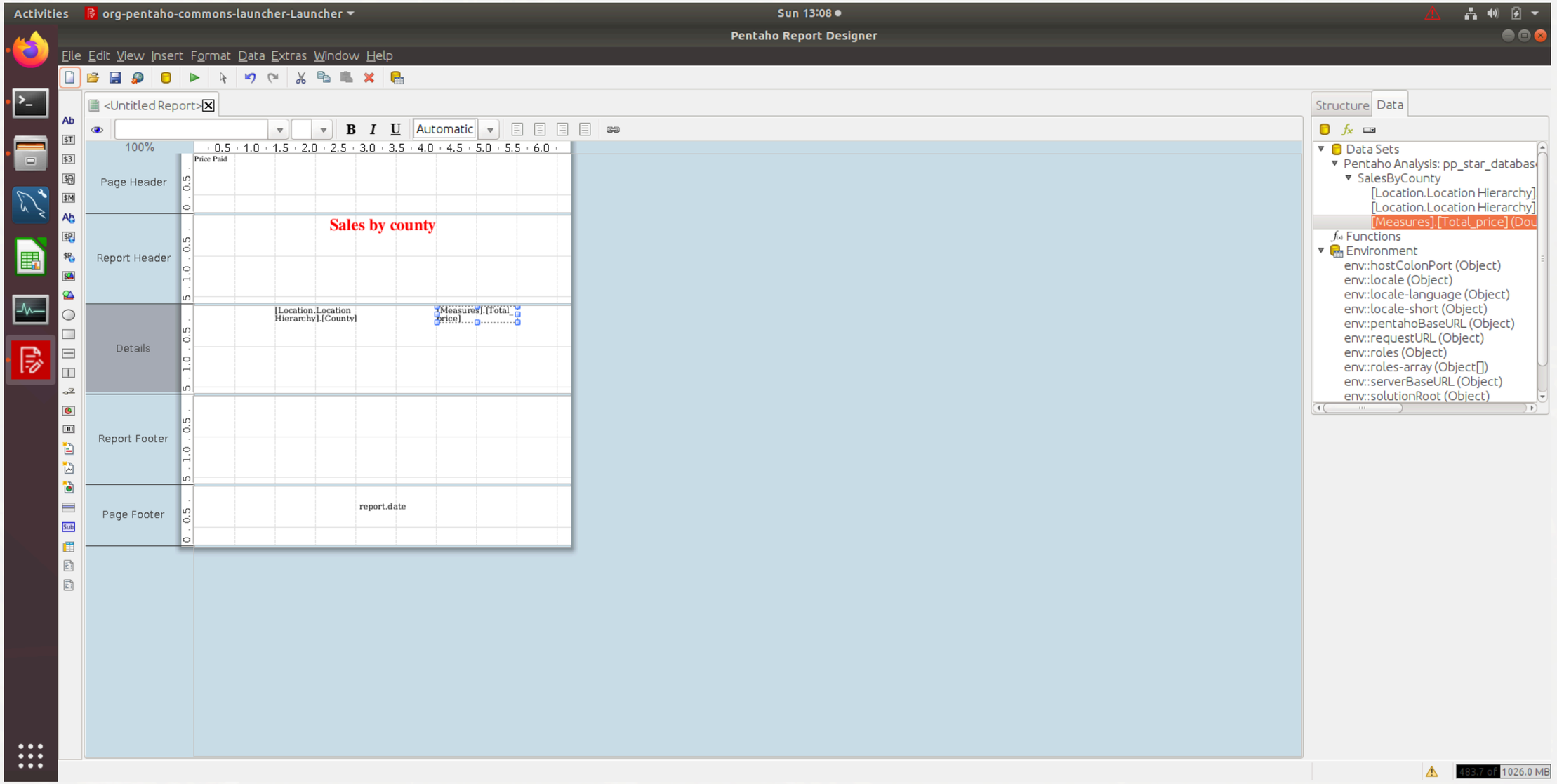


County	District	Jan	Feb	Mar	Apr	May	Jun	July	Aug
GREATER LONDON		4,852,026,294	4,656,116,852	8,000,674,940	1,523,582,590	2,382,698,101	2,491,036,553	2,773,898,445	1,303,754,890
	BARKING AND DAGENHAM	40,350,250	44,027,000	81,826,900	13,342,995	23,407,200	16,358,000	33,591,634	9,302,750
	BARNET	168,337,508	180,217,048	361,463,697	64,285,890	118,892,660	109,789,226	114,349,753	68,879,050
	BEXLEY	120,323,800	113,268,446	198,731,914	30,221,835	44,245,400	55,105,050	63,586,155	26,722,745
	BRENT	115,001,478	101,159,215	179,302,836	26,083,950	51,141,231	53,344,476	55,604,479	23,280,125
	BROMLEY	185,888,612	230,253,739	376,457,878	51,687,859	102,305,749	124,217,946	140,388,631	65,121,132
	CAMDEN	263,703,715	160,595,096	251,093,678	78,681,630	79,416,634	102,399,225	81,484,418	41,120,180
	CITY OF LONDON	123,577,395	17,216,300	22,314,999	66,860,000	9,385,000	3,870,000	5,083,000	2,375,000
	CITY OF WESTMINSTER	456,541,049	233,243,574	496,096,529	127,407,500	161,545,745	210,143,359	72,249,779	64,519,168
	CROYDON	127,773,152	127,898,304	275,127,712	55,103,450	91,701,995	99,874,599	103,458,115	51,211,965
	EALING	156,670,046	156,472,409	232,452,574	48,155,495	82,656,450	64,607,884	101,358,594	47,996,952
	ENFIELD	108,516,156	99,057,990	213,676,222	37,555,495	59,494,116	57,374,545	69,108,380	41,411,392
	GREENWICH	110,654,647	91,609,892	209,915,531	32,565,046	60,103,159	68,929,440	71,344,696	28,359,445
	HACKNEY	117,459,572	116,578,031	228,410,865	29,741,242	51,459,378	73,531,857	75,419,054	33,770,150
	HAMMERSMITH AND FULHAM	194,059,912	150,931,823	240,563,416	67,099,744	72,303,248	72,715,200	100,830,296	50,315,505
	HARINGEY	138,419,271	164,694,230	259,508,471	43,539,220	47,920,883	85,499,601	87,274,948	43,162,070
	HARROW	68,837,906	80,701,251	166,315,717	21,105,450	53,916,295	36,869,600	50,864,133	24,652,810
	HAVERING	121,209,554	144,067,408	231,956,251	29,659,800	61,337,150	64,634,217	89,587,500	39,249,500
	HILLINGDON	98,881,767	103,736,788	218,479,264	24,378,950	58,959,268	71,939,451	54,547,399	35,051,450
	HOUNSLOW	88,455,792	99,985,586	186,778,600	37,318,892	43,344,444	51,693,036	68,479,818	22,905,006
	ISLINGTON	140,690,740	108,513,148	213,577,130	40,024,357	55,329,530	71,401,216	86,286,759	29,576,500
	KENSINGTON AND CHELSEA	276,687,484	377,574,578	264,160,581	110,968,318	125,850,261	105,673,650	136,680,625	41,751,731
	KINGSTON UPON THAMES	97,344,465	96,059,855	190,452,067	36,532,775	62,874,096	48,545,675	65,914,633	30,538,698
	LAMBETH	155,546,041	164,372,163	332,087,875	36,570,684	91,755,752	91,177,300	114,197,514	71,352,913
	LEWISHAM	154,337,648	161,964,679	246,153,039	39,757,350	69,281,245	80,722,419	96,085,998	41,249,071
	MERTON	133,622,849	180,392,891	251,632,074	42,972,333	98,279,900	60,404,958	102,684,265	35,231,320

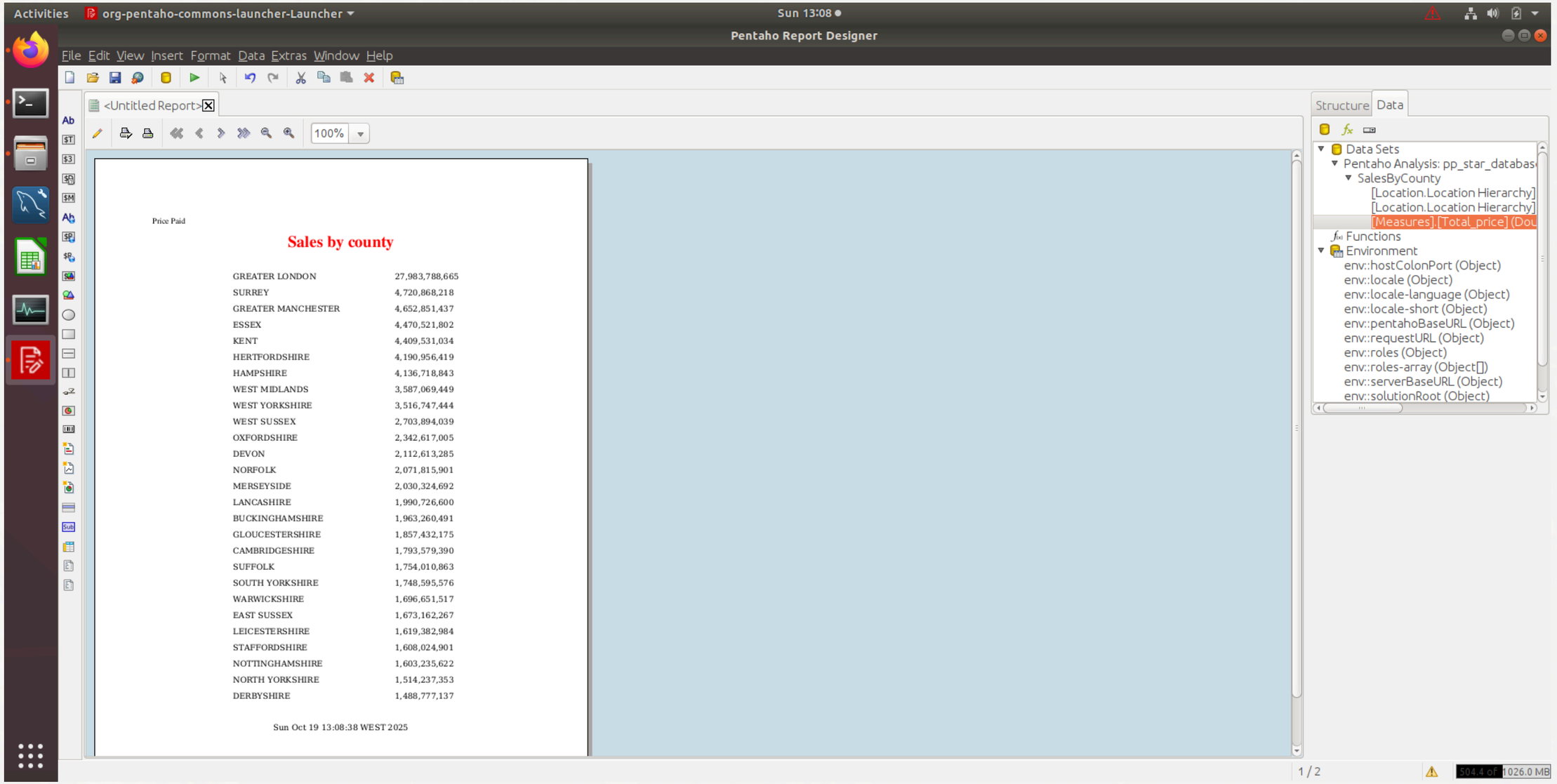
TASK 10 - MDX Query



TASK 10 - Report (design mode)



TASK 10 - Report (preview mode)



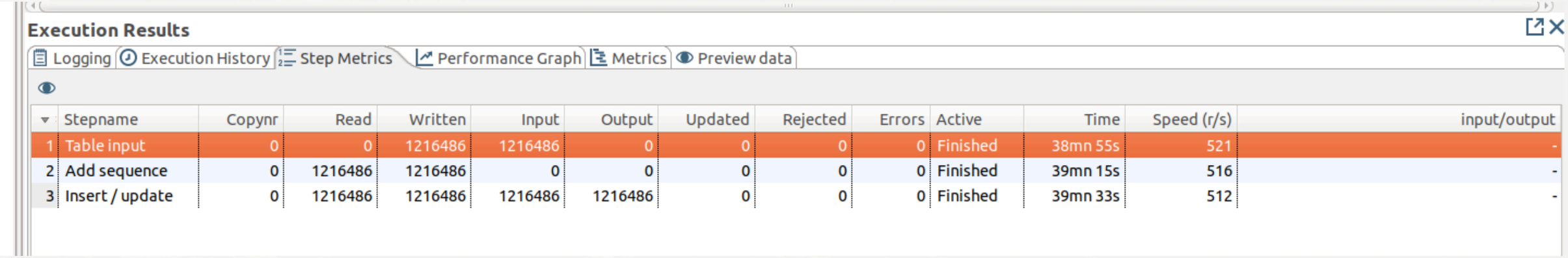
TASK 11 -Loading the 2024 data

```
LOAD DATA LOCAL INFILE '/home/aid/Downloads/pp-2024.csv'
INTO TABLE pp_table
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n';
```

TASK 11 - Bottlenecks

Task 5:

- On task 5's transformation (loading data into the location dimension), the main bottleneck is the Table Input step. This is because we query multiple (8) values from the original database and ask only for DISTINCT combinations.
- For each fact in the Fact Table to represent a single sale, we needed each “locationid” to represent the location of a single property. For this, it was necessary to include not only County, District, and City, but also more granular columns like postcode, that are necessary to identify a specific property. “Distinct” was necessary because of the “Add Sequence” step that followed - each locationid had to represent a distinct property’s location.



	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	Table input	0	0	1216486	1216486	0	0	0	0	Finished	38mn 55s	521	-
2	Add sequence	0	1216486	1216486	0	0	0	0	0	Finished	39mn 15s	516	-
3	Insert / update	0	1216486	1216486	1216486	1216486	0	0	0	Finished	39mn 33s	512	-

TASK 11 - Bottlenecks

Task 6:

- On task 6's transformation (loading data into the time dimension), we could identify no bottlenecks. All tasks finished approximately at the same time.
- We also use “DISTINCT” for task 6's query, but we only query for a single column (transferdate) and with two years worth of data there were less than a thousand rows.

TASK 11 - Bottlenecks

Task 7:

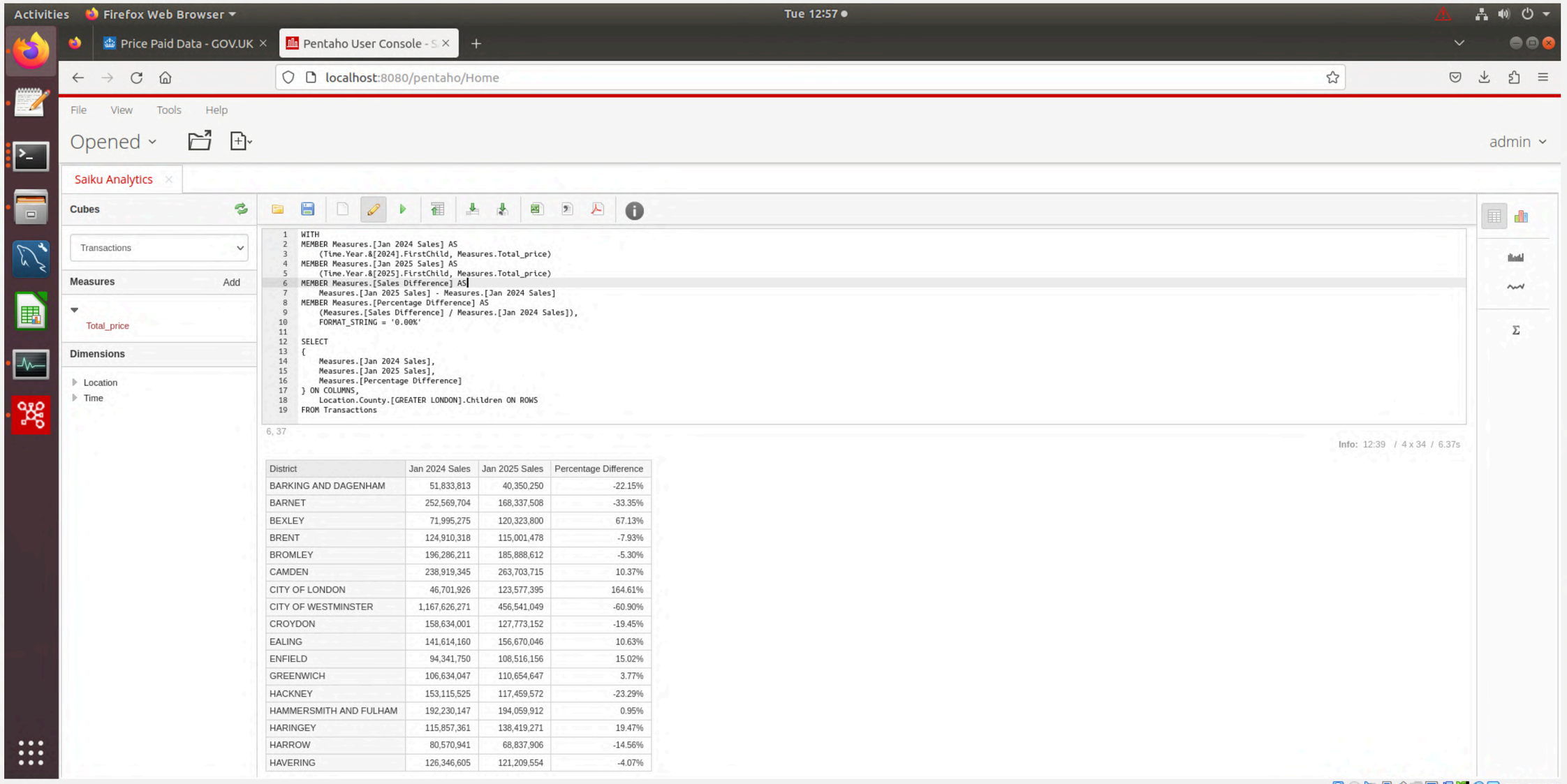
- On task 7's transformation (loading data into the fact table), the main bottleneck is the Sort Rows step. This is because Sort Rows is a blocking step: it needs to read all rows from the input before it can start outputting rows to the next step (Group By).

Execution Results

Logging | Execution History | Step Metrics | Performance Graph | Metrics | Preview data

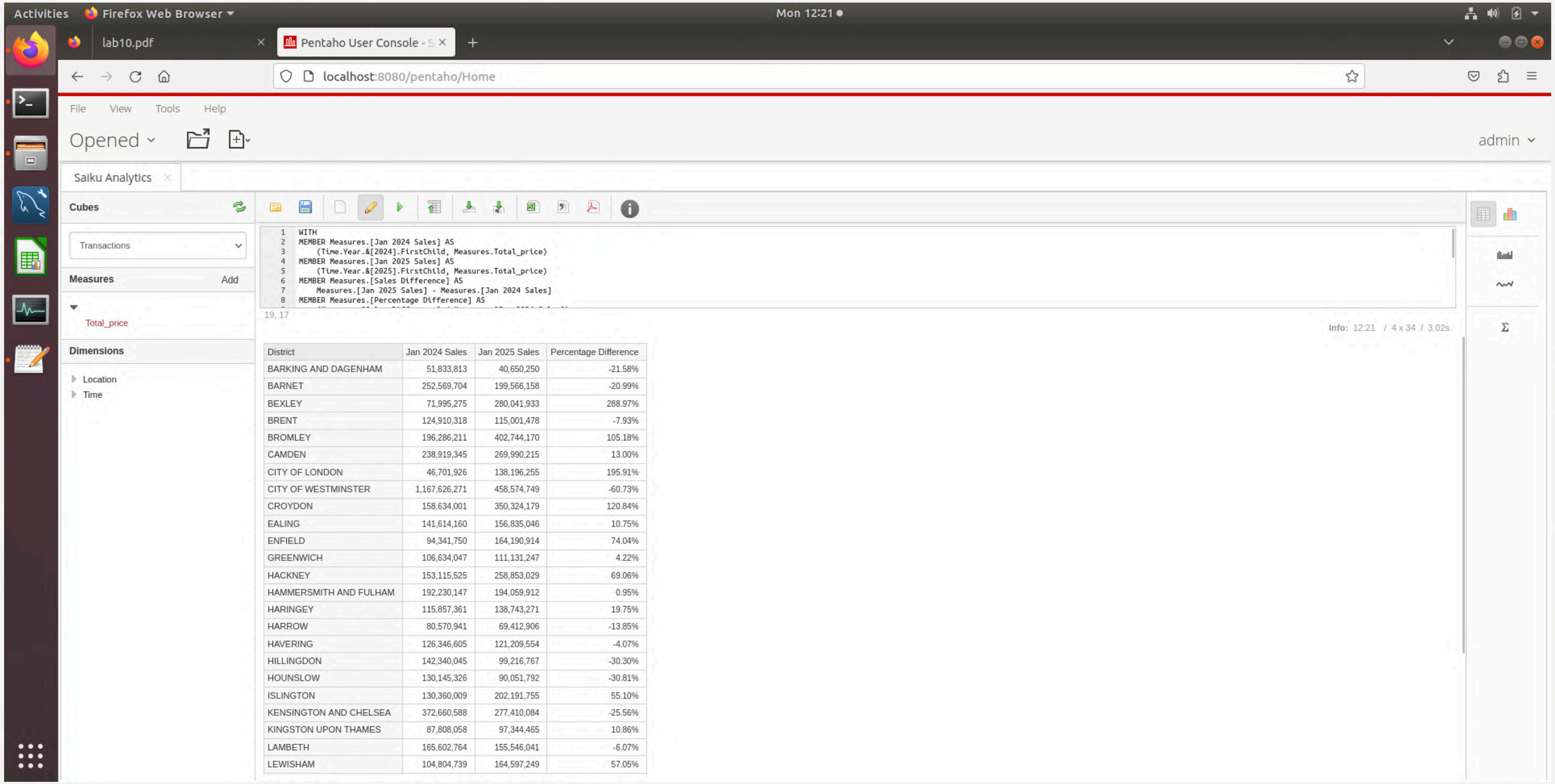
	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	Table input 2	0	0	1216486	1216486	0	0	0	0	Finished	55.9s	21,757	-
2	Table input	0	0	1236593	1236593	0	0	0	0	Finished	1mn 56s	10,581	-
3	If field value is null	0	1216486	1216486	0	0	0	0	0	Finished	55.9s	21,756	-
4	If field value is null 2	0	1236593	1236593	0	0	0	0	0	Finished	1mn 56s	10,581	-
5	Stream lookup	0	2453079	1236593	0	0	0	0	0	Finished	1mn 56s	20,989	-
6	Sort rows	0	1236593	1236593	0	0	0	0	0	Finished	38mn 56s	529	-
7	Group by	0	1236593	1231776	0	0	0	0	0	Finished	39mn 13s	526	-
8	Insert / update	0	1231776	1231776	1231776	1231776	0	0	0	Finished	39mn 28s	520	-

TASK 12 - MDX Query



District	Jan 2024 Sales	Jan 2025 Sales	Percentage Difference
BARKING AND DAGENHAM	51,833,813	40,350,250	-22.15%
BARNET	252,569,704	168,337,508	-33.35%
BEXLEY	71,995,275	120,323,800	67.13%
BRENT	124,910,318	115,001,478	-7.93%
BROMLEY	196,286,211	185,888,612	-5.30%
CAMDEN	238,919,345	263,703,715	10.37%
CITY OF LONDON	46,701,926	123,577,395	164.61%
CITY OF WESTMINSTER	1,167,626,271	456,541,049	-60.90%
CROYDON	158,634,001	127,773,152	-19.45%
EALING	141,614,160	156,670,046	10.63%
ENFIELD	94,341,750	108,516,156	15.02%
GREENWICH	106,634,047	110,654,647	3.77%
HACKNEY	153,115,525	117,459,572	-23.29%
HAMMERSMITH AND FULHAM	192,230,147	194,059,912	0.95%
HARINGEY	115,857,361	138,419,271	19.47%
HARROW	80,570,941	68,837,906	-14.56%
HAVERING	126,346,605	121,209,554	-4.07%

TASK 12 - MDX Query results



District	Jan 2024 Sales	Jan 2025 Sales	Percentage Difference
BARKING AND DAGENHAM	51,833,813	40,650,250	-21.58%
BARNET	252,569,704	199,566,158	-20.99%
BEXLEY	71,995,275	280,041,933	288.97%
BRENT	124,910,318	115,001,478	-7.93%
BROMLEY	196,286,211	402,744,170	105.18%
CAMDEN	238,919,345	269,990,215	13.00%
CITY OF LONDON	46,701,926	138,196,255	195.91%
CITY OF WESTMINSTER	1,167,626,271	458,574,749	-60.73%
CROYDON	158,634,001	350,324,179	120.84%
EALING	141,614,160	156,835,046	10.75%
ENFIELD	94,341,750	164,190,914	74.04%
GREENWICH	106,634,047	111,131,247	4.22%
HACKNEY	153,115,525	258,853,029	69.06%
HAMMERSMITH AND FULHAM	192,230,147	194,059,912	0.95%
HARINGEY	115,857,361	138,743,271	19.75%
HARROW	80,570,941	69,412,906	-13.85%
HAVERING	126,346,605	121,209,554	-4.07%
HILLINGDON	142,340,045	99,216,767	-30.30%
HOUNSLOW	130,145,326	90,051,792	-30.81%
ISLINGTON	130,360,009	202,191,755	55.10%
KENSINGTON AND CHELSEA	372,660,588	277,410,084	-25.56%
KINGSTON UPON THAMES	87,808,058	97,344,465	10.86%
LAMBETH	165,602,764	155,546,041	-6.07%
LEWISHAM	104,804,739	164,597,249	57.05%